

Intermediate Microeconomic

Part two: Choice and Demand

Week 2(b): Utility Maximization and Choice

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Consumer Theory

- A consumer wakes up in the morning. The world has two goods (commodities) for consumption: x and y . He looks outside the window and observe prices P_x and P_y . He reaches into his pocket and find M dollars there; he knows his preferences over goods x and y .
- Consumer's problem: find the best affordable combination of goods x and y for him to eat
- Two building blocks of the consumer theory:
 - **Preference**
 - **Budget constraint**

Optimization principle

- To maximize utility, given a fixed amount of income to spend, an individual will buy the goods and services:
 - that exhaust his or her total income
 - for which the rate of trade-off between any goods (the *MRS*) is equal to the rate at which goods can be traded for one another in the marketplace
-

A numerical illustration

- Assume that the individual's $MRS = 1$
 - What does that imply?

Budget Constraint

- Consumers' ability to consume is limited by what they can afford. Let p_x be the price of x and p_y be the price of y , and suppose the consumer has m units of wealth to spend on x and y .

- Then the consumer's choice must satisfy the ***budget constraint***:

$$p_x x + p_y y \leq m$$

- If consumers maximize their utility, there is no satiation and no savings, then the budget constraint should hold with equality. So we usually write budget constraint as:

$$p_x x + p_y y = m$$

Budget Constraint

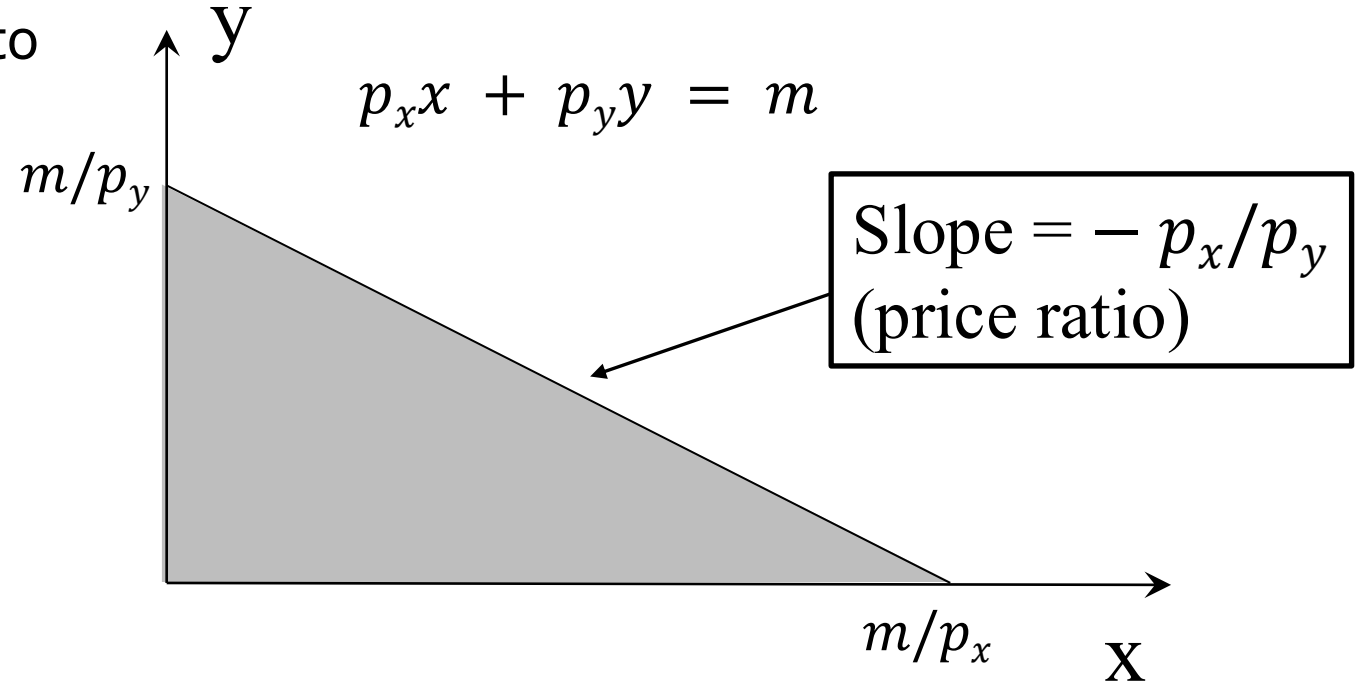
- ☐ **Question: Cookies (x) cost \$1 each and milk (y) costs \$2.5 per gallon. You have \$10 to spend. Write down your budget constraint.**

Graphing budget constraint

- Rearrange the budget equation to get:

$$y = -p_x/p_y x + m/p_y$$

- Slope is $-p_x/p_y$.
- Spend all m on x , can buy m/p_x units.
- Spend all m on y , can buy m/p_y units.
- Set of **feasible choices** in gray.



What if m increases from m_1 to m_2 ?

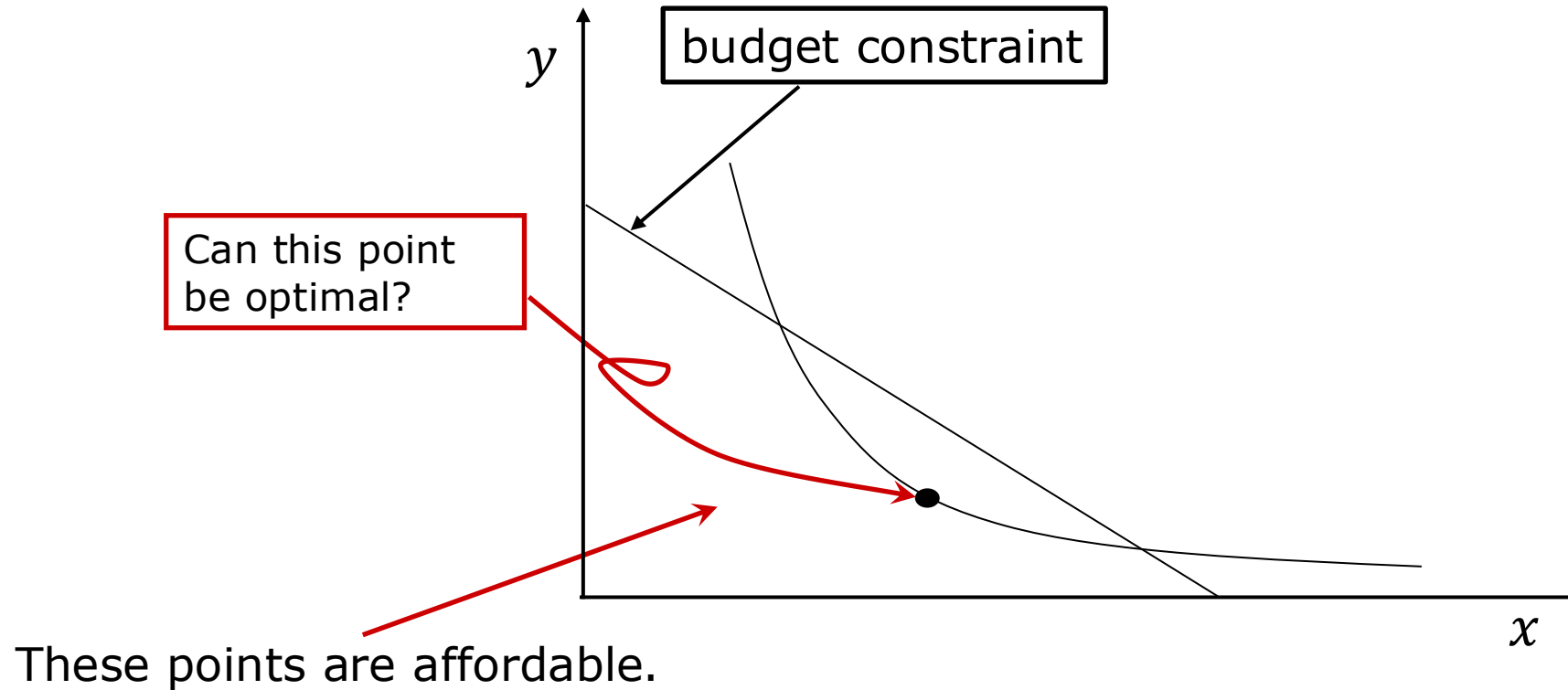
What if p_x increases from p_{x1} to p_{x2} ?

Utility Maximization Problem

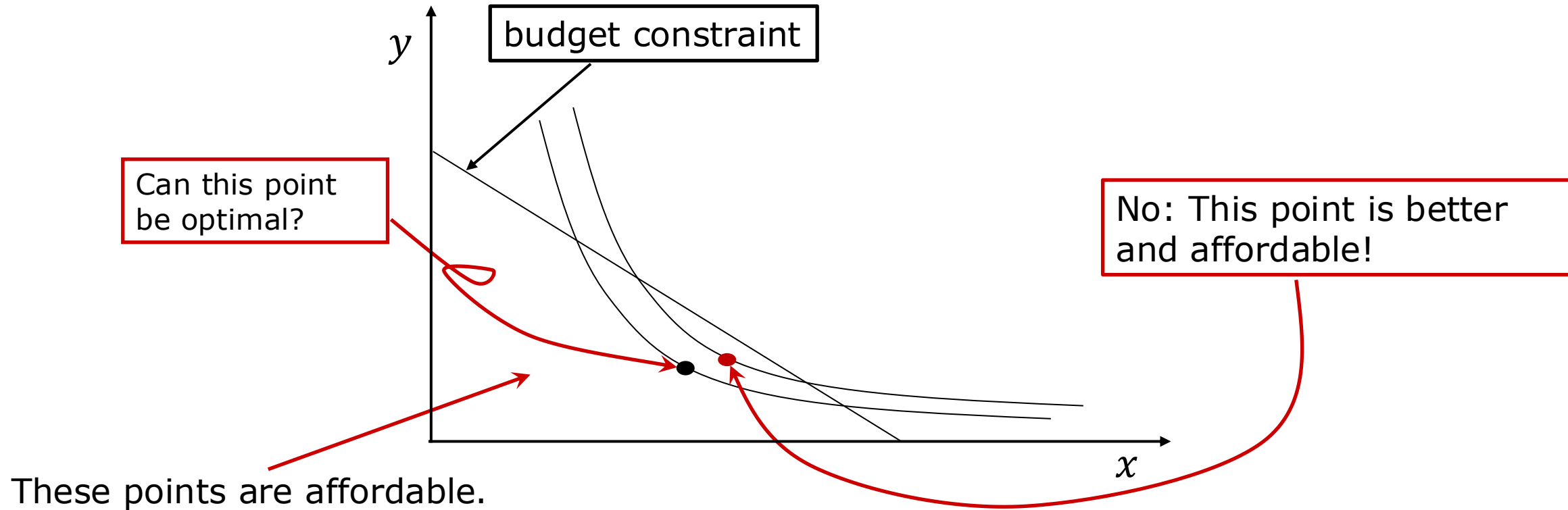
- The consumer has preferences represented by a utility function.
- The consumption bundles available are those that satisfy the budget constraint.
- Given this constraint, the consumer chooses the bundle that maximizes his utility function.
- We call this the consumer's **Utility Maximization Problem (UMP)**:

$$\begin{array}{l} \max_{x,y} u(x,y) \\ \text{subject to: } p_x x + p_y y \leq m \\ x, y \geq 0 \end{array}$$

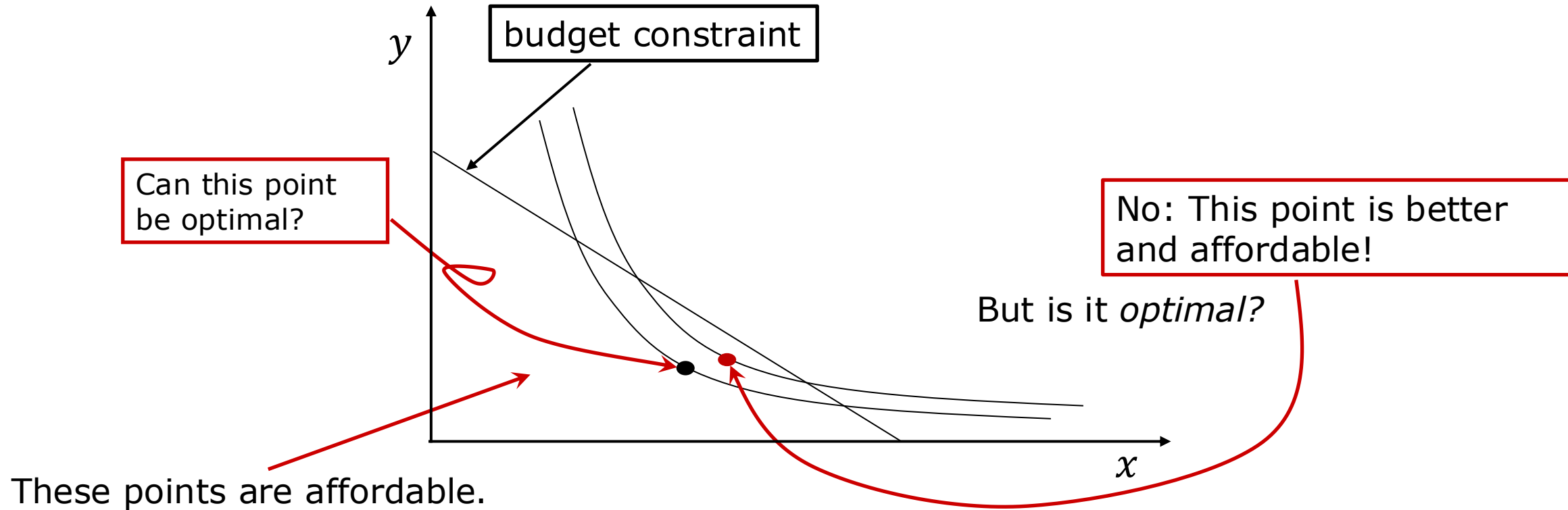
Solving the UMP Graphically



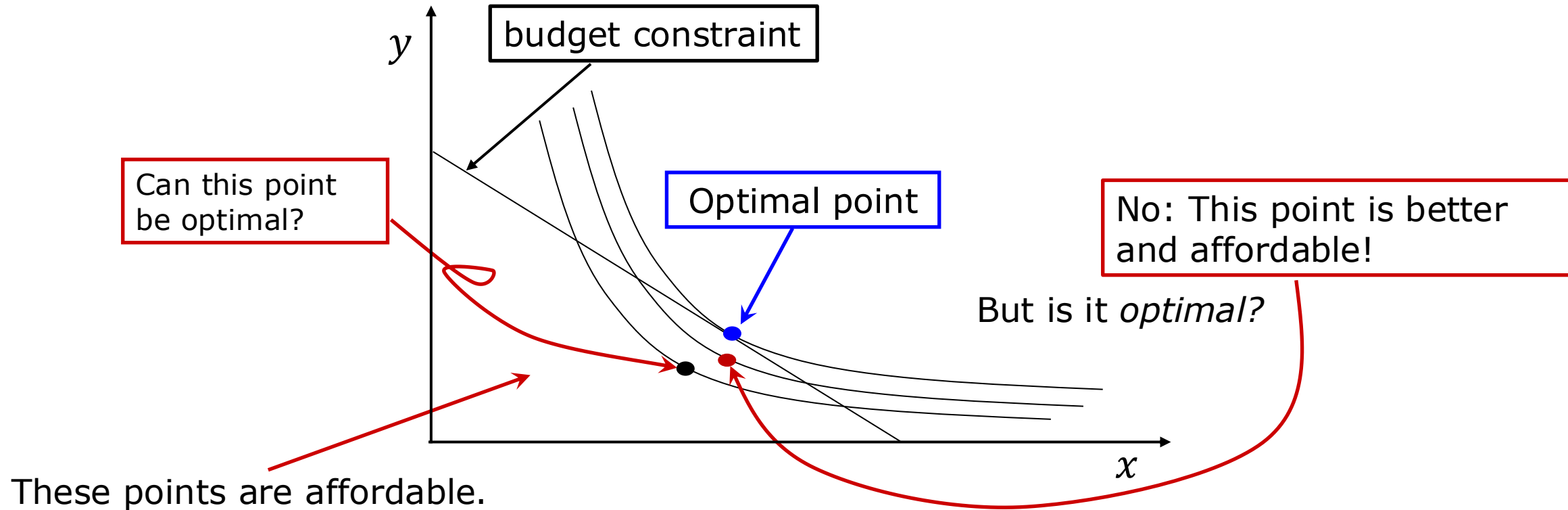
Solving the UMP Graphically



Solving the UMP Graphically



Solving the UMP Graphically



- The optimal point is one where there is no affordable point that is better. This is a point of **tangency** between the budget line and an indifference curve.

Solving the UMP algebraically

□ This means that the following two conditions determine the optimal point (x^*, y^*) :

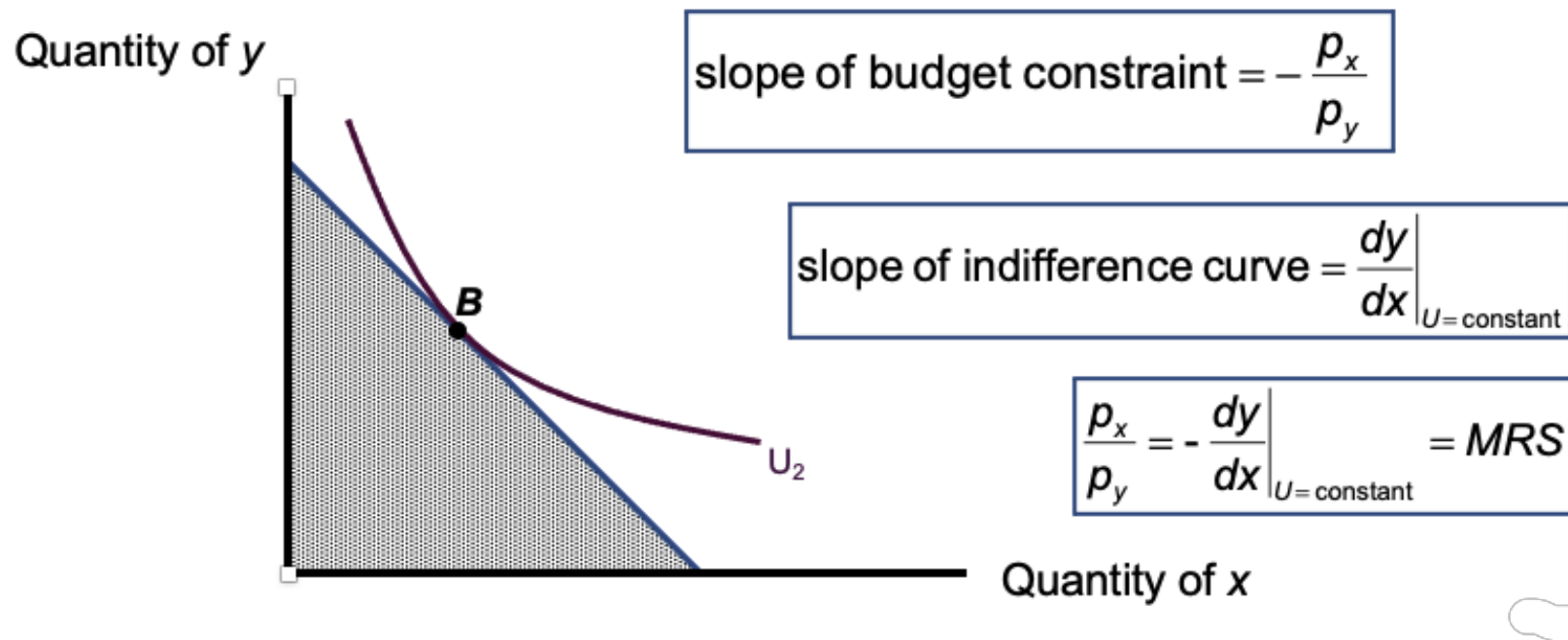
1. (x^*, y^*) is on the budget line: $p_x x^* + p_y y^* = m$.

2. The |slope of the budget line| = |the slope of the indifference curve| (which is |MRS|):

$$MRS = \frac{\frac{\partial U(x^*, y^*)}{\partial x}}{\frac{\partial U(x^*, y^*)}{\partial y}} = \frac{p_x}{p_y}$$

First-order conditions for a maximum

- Utility is maximized where the indifference curve is tangent to the budget constraint



Interpreting the tangency condition

□ Let U_x denote $\partial U / \partial x$ and U_y denote $\partial U / \partial y$. (In general, $U_n = \partial U / \partial n$.)

□ Rearrange the tangency condition to get: $\frac{U_x}{p_x} = \frac{U_y}{p_y}$.

□ Note $\frac{U_x}{p_x} = U_x \cdot \frac{1}{p_x}$ → How much x you can buy if you spend 1 more dollar on x

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How much x you can buy if you spend 1 more dollar on x

How much utility you get if you consumer a little more of x

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How much x you can buy if you spend 1 more dollar on x

How much utility you get if you consumer a little more of x

□ So U_x / p_x is the additional utility from spending another dollar on x .

Interpreting the tangency condition

□ $\frac{U_x}{p_x} = \frac{U_y}{p_y}$

- At the optimum, the additional utility from spending another dollar on x must equal the additional utility from spending another dollar on y .
- Why must this be?
- Question: What happens if $U_x/p_x > U_y/p_y$? What should the consumer do?

Solving the UMP algebraically: example

□ Consider $U(x, y) = xy$

□ Step 1. Invoke optimal condition:

$$\frac{U_x}{p_x} = \frac{U_y}{p_y}$$

$U_x = \partial U / \partial x = y$, $U_y = \partial U / \partial y = x$, so

$$\frac{y^*}{p_x} = \frac{x^*}{p_y}$$

Rearranging this gives:

$$y^* = \frac{p_x x^*}{p_y}$$

Solving the UMP algebraically: example

- Step 2: Substitute $y^* = \frac{p_x x^*}{p_y}$ into budget constraint:

$$p_x x^* + p_y y^* = m$$

$$p_x x^* + p_y \frac{p_x x^*}{p_y} = m$$

$$2p_x x^* = m$$

Demand for x

$$x^* = \frac{m}{2p_x}$$

Demand for y

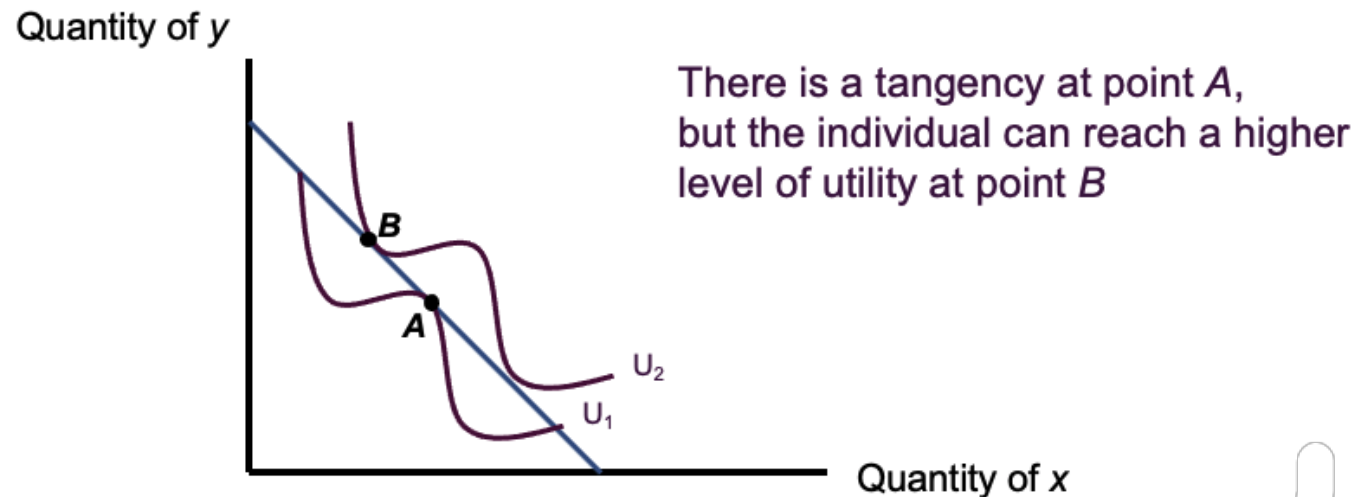
$$y^* = \frac{p_x x^*}{p_y} = \frac{m}{2p_y}$$

- Note that the solutions are **functions** of prices (p_x, p_y) and wealth (m).
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Second-order conditions for a maximum

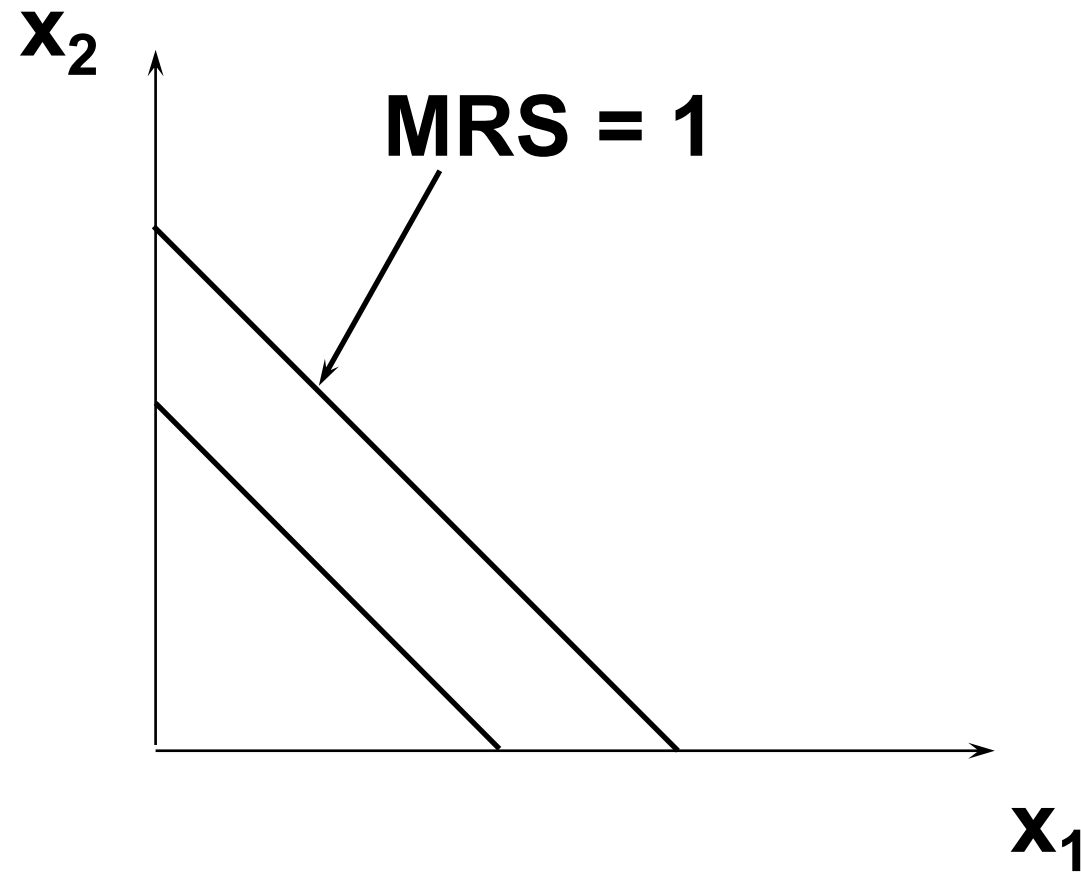
- The tangency rule is only necessary but not sufficient
 - We need MRS to be diminishing
 - then indifference curves are strictly convex

- If MRS is not diminishing, then we must check second-order conditions to ensure that we are at a maximum



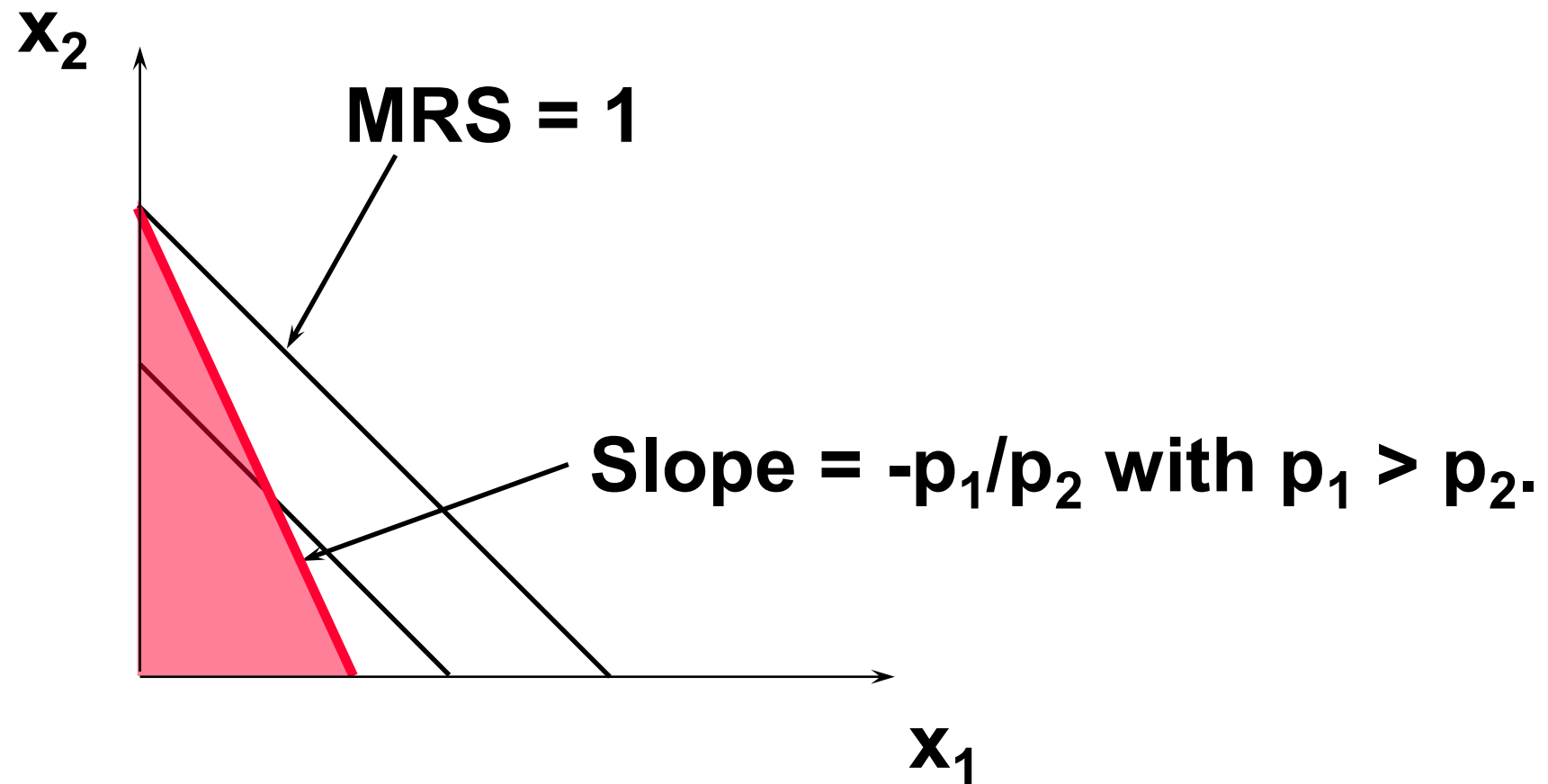
Examples of Corner Solutions

-- the Perfect Substitutes Case



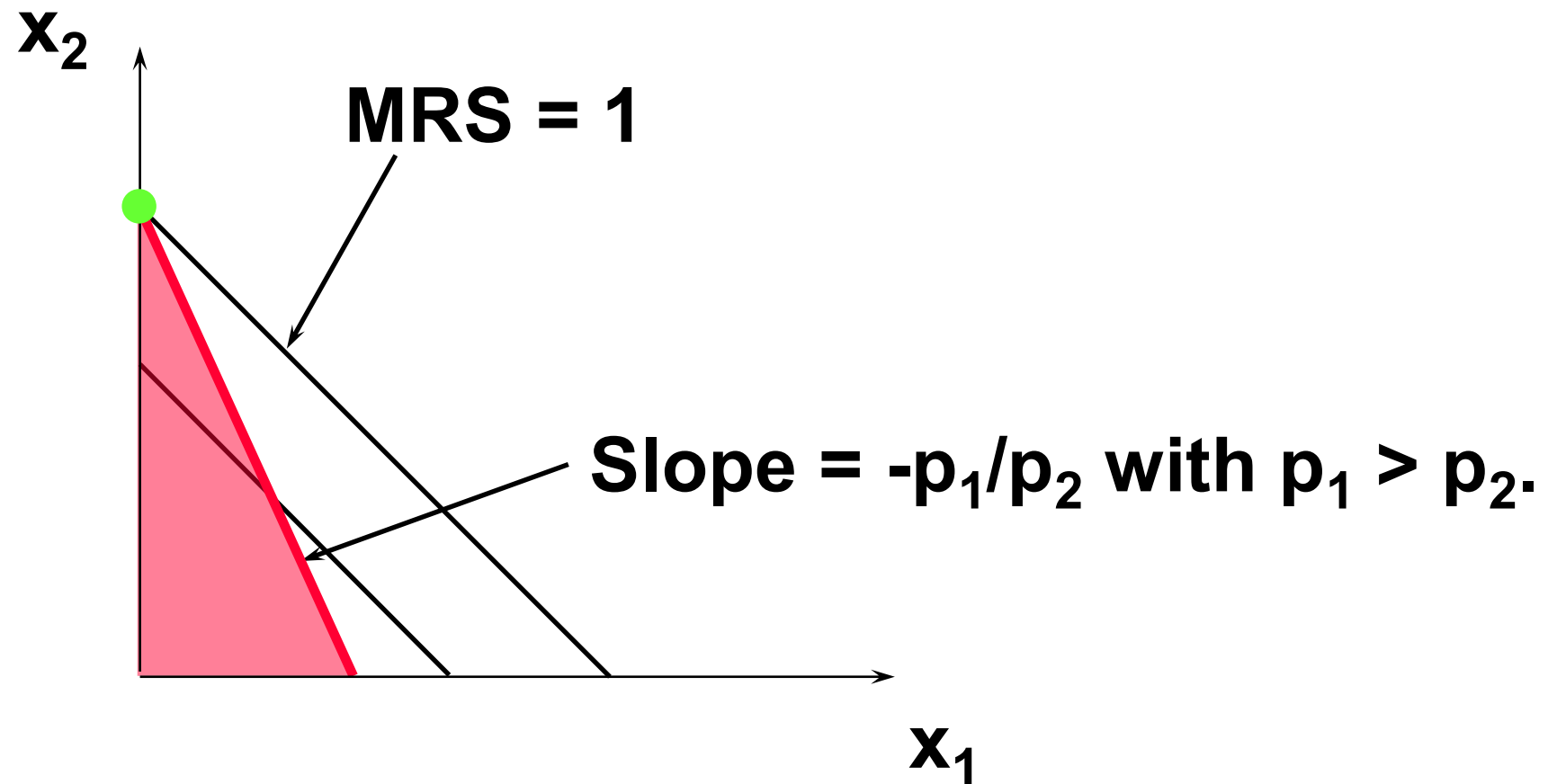
Examples of Corner Solutions

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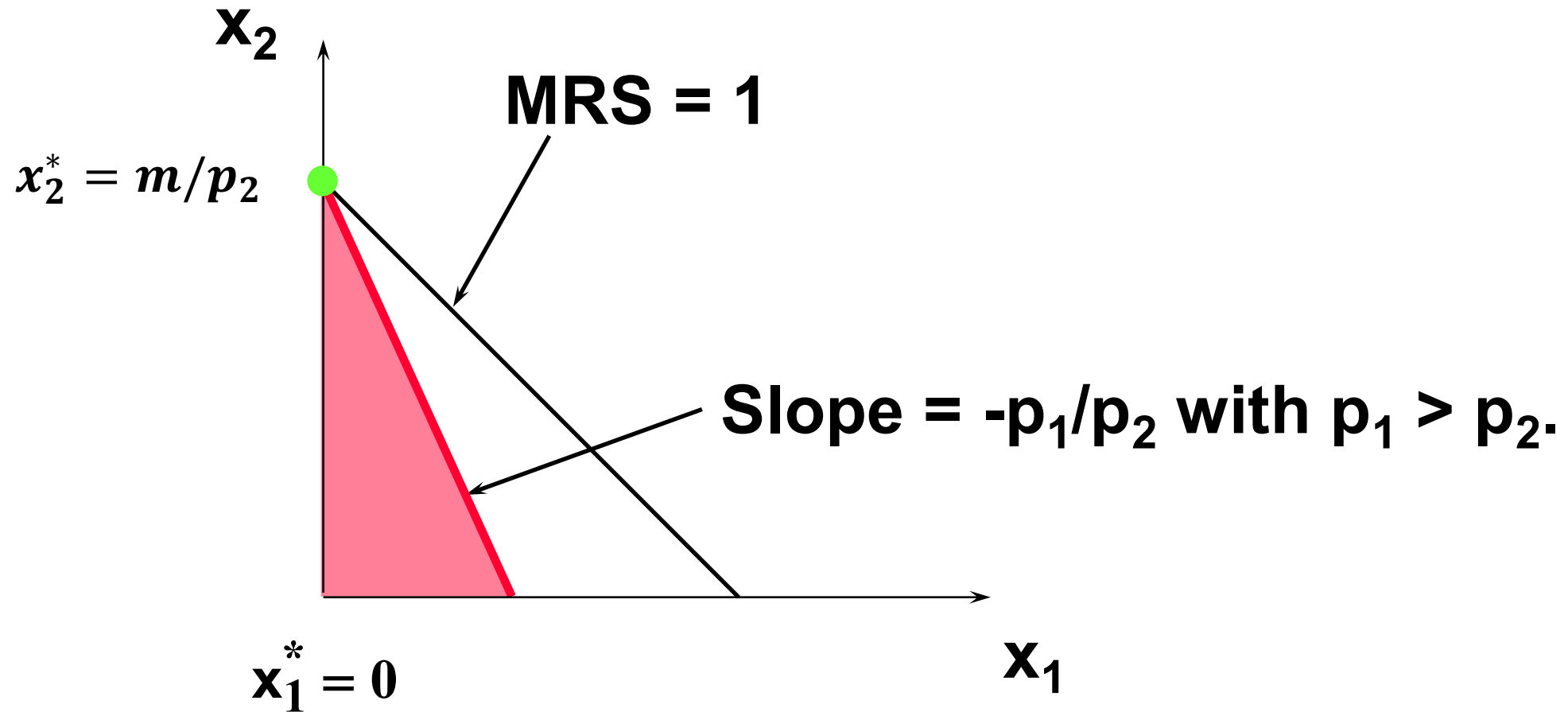
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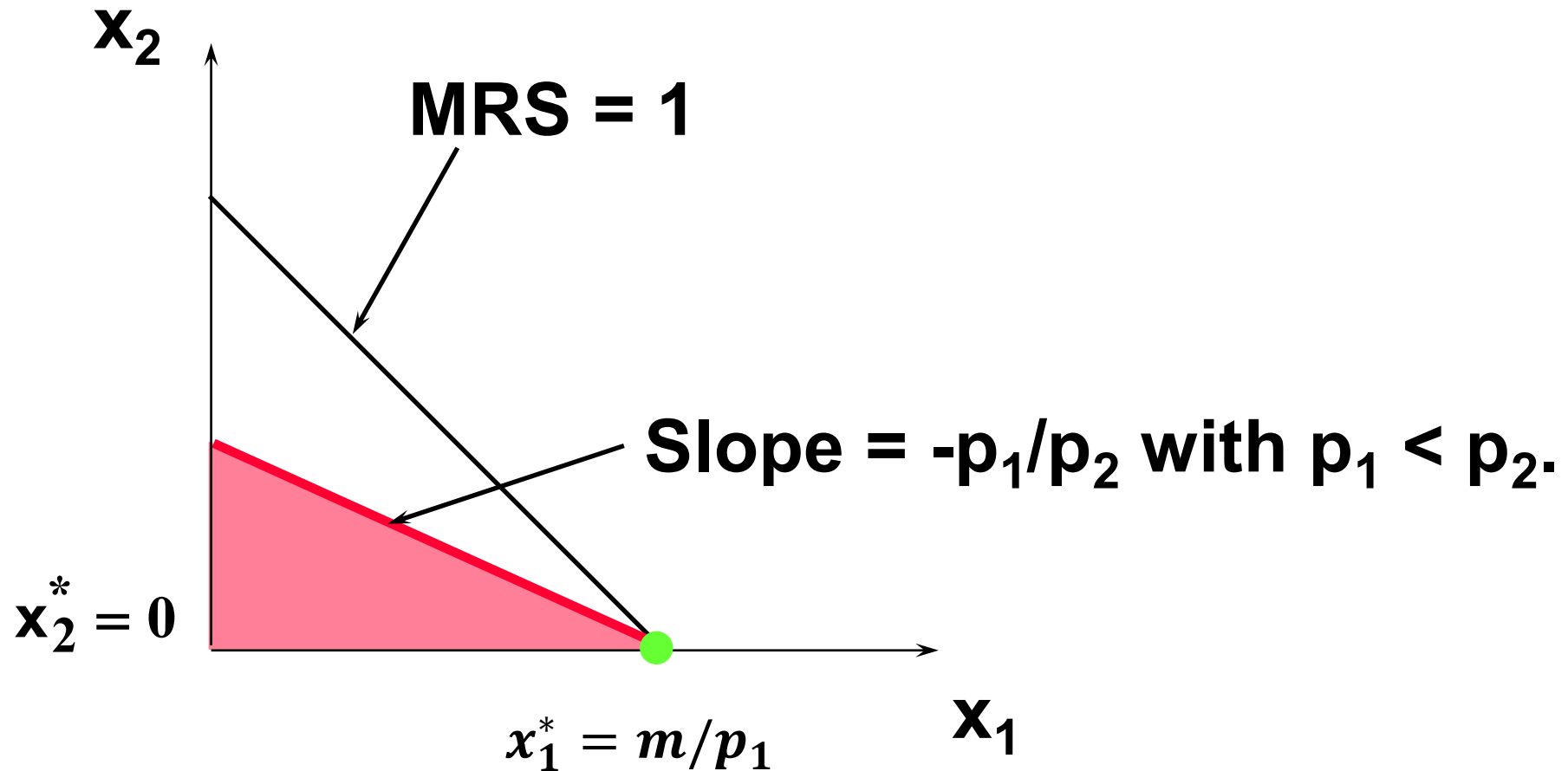
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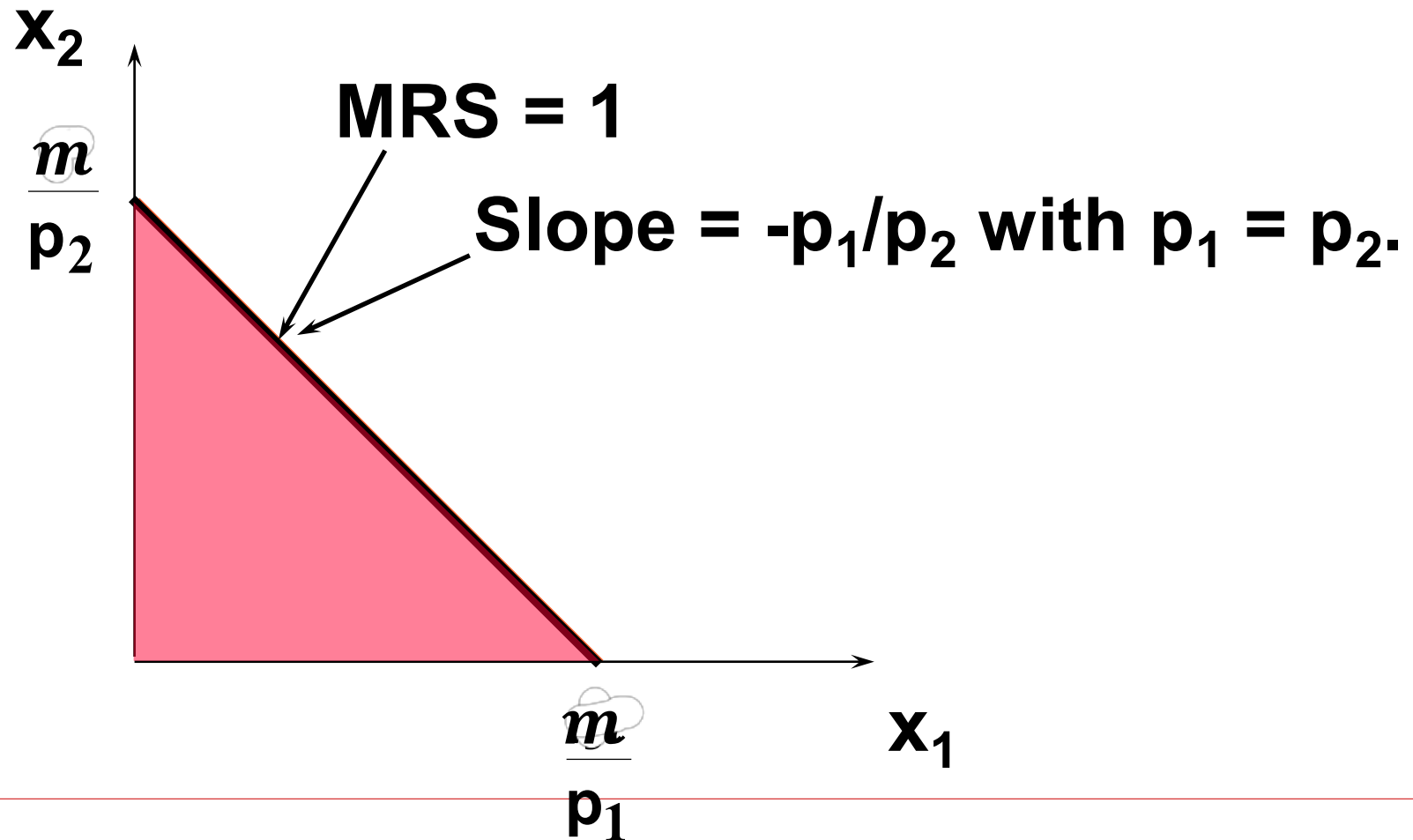
So when $U(x_1, x_2) = x_1 + x_2$, the most preferred affordable bundle is (x_1^*, x_2^*) where

and
$$(x_1^*, x_2^*) = \left(\frac{m}{p_1}, 0 \right) \quad \text{if } p_1 < p_2$$

$$(x_1^*, x_2^*) = \left(0, \frac{m}{p_2} \right) \quad \text{if } p_1 > p_2.$$

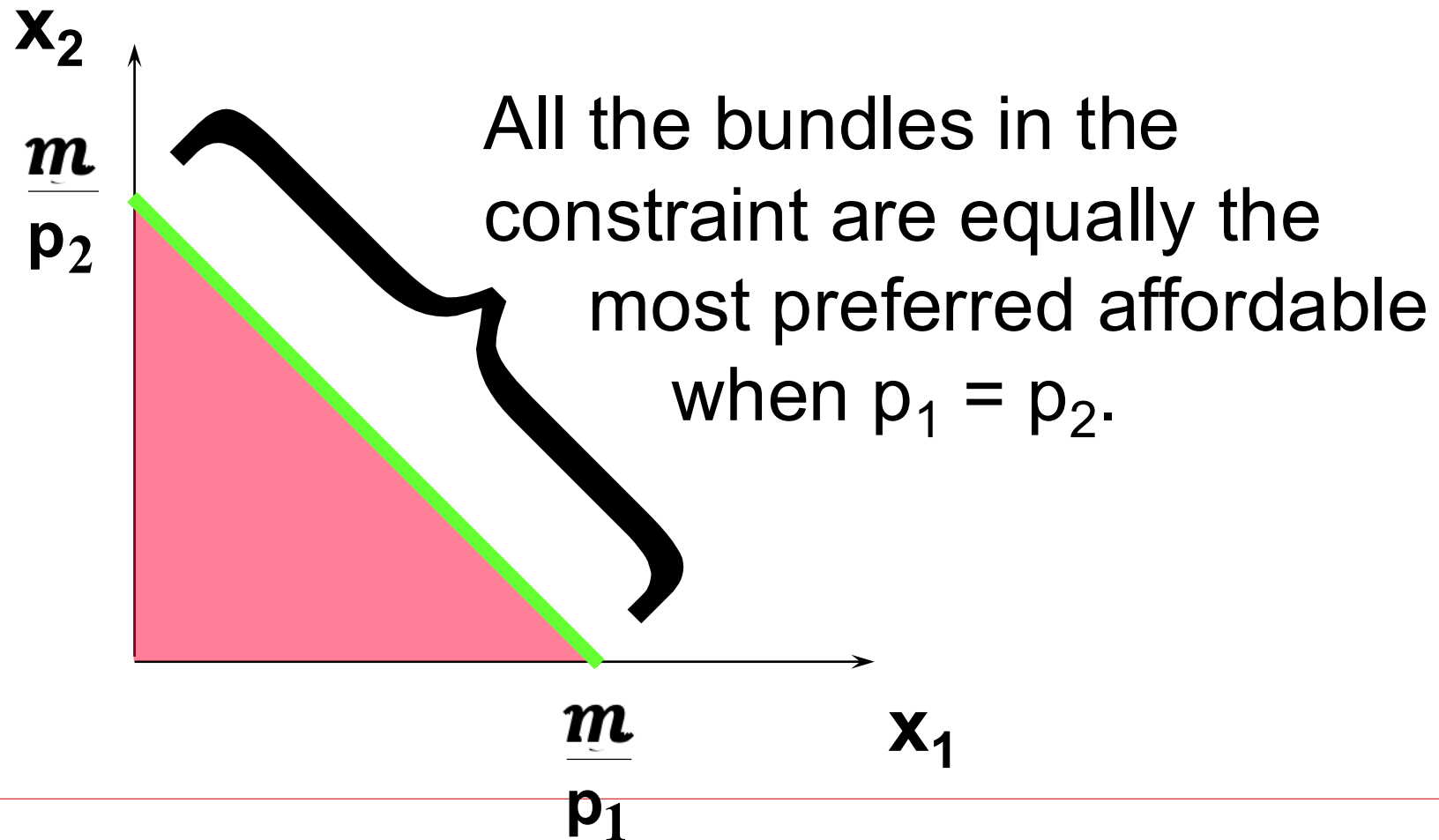
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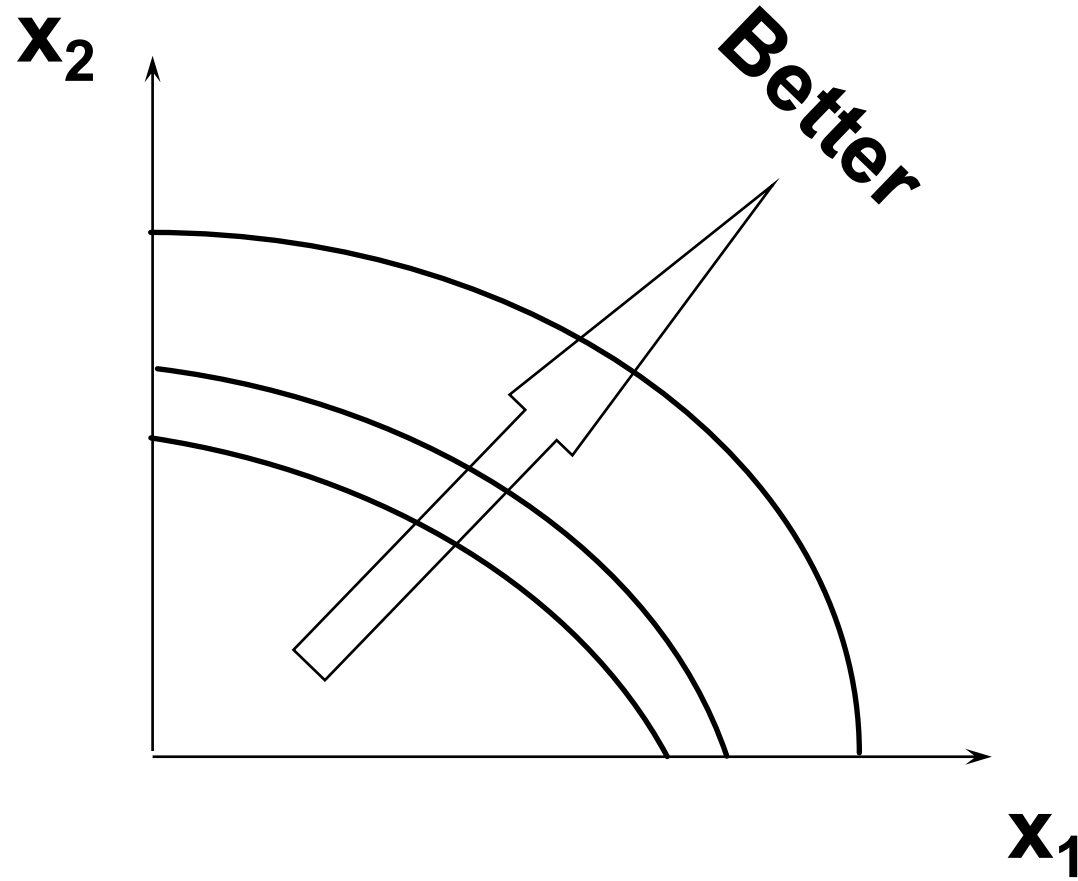
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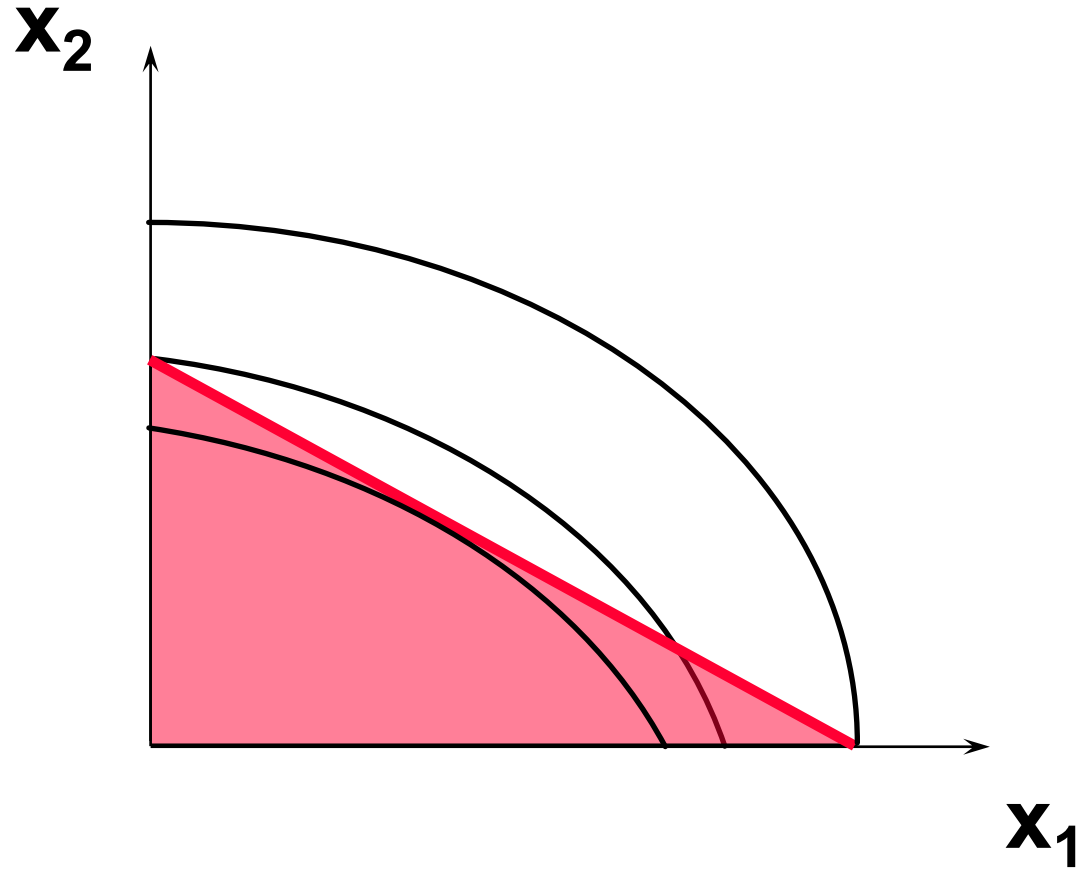
Examples of Corner Solutions

-- the Non-Convex Preferences Case



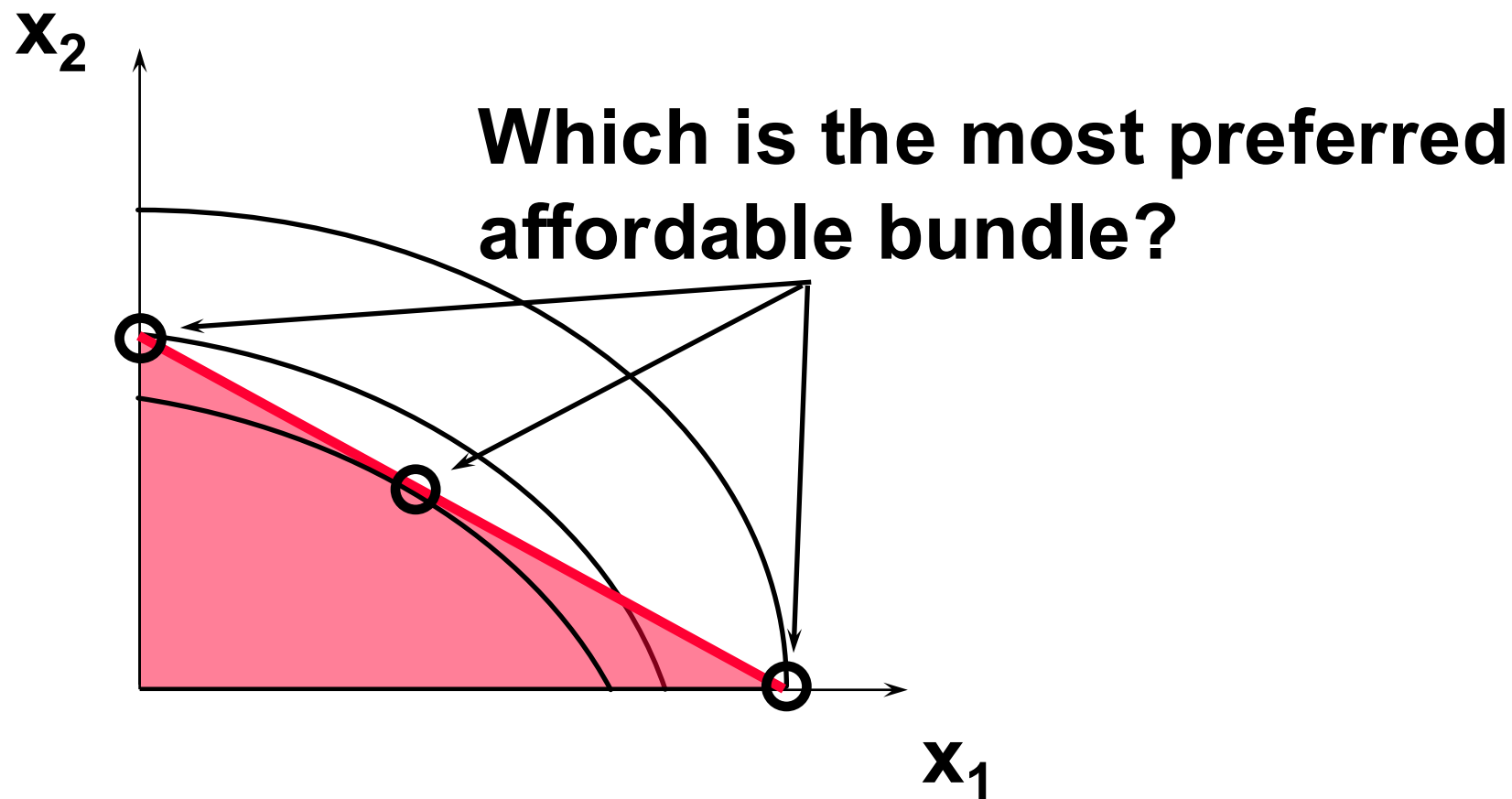
Examples of Corner Solutions

-- the Non-Convex Preferences Case



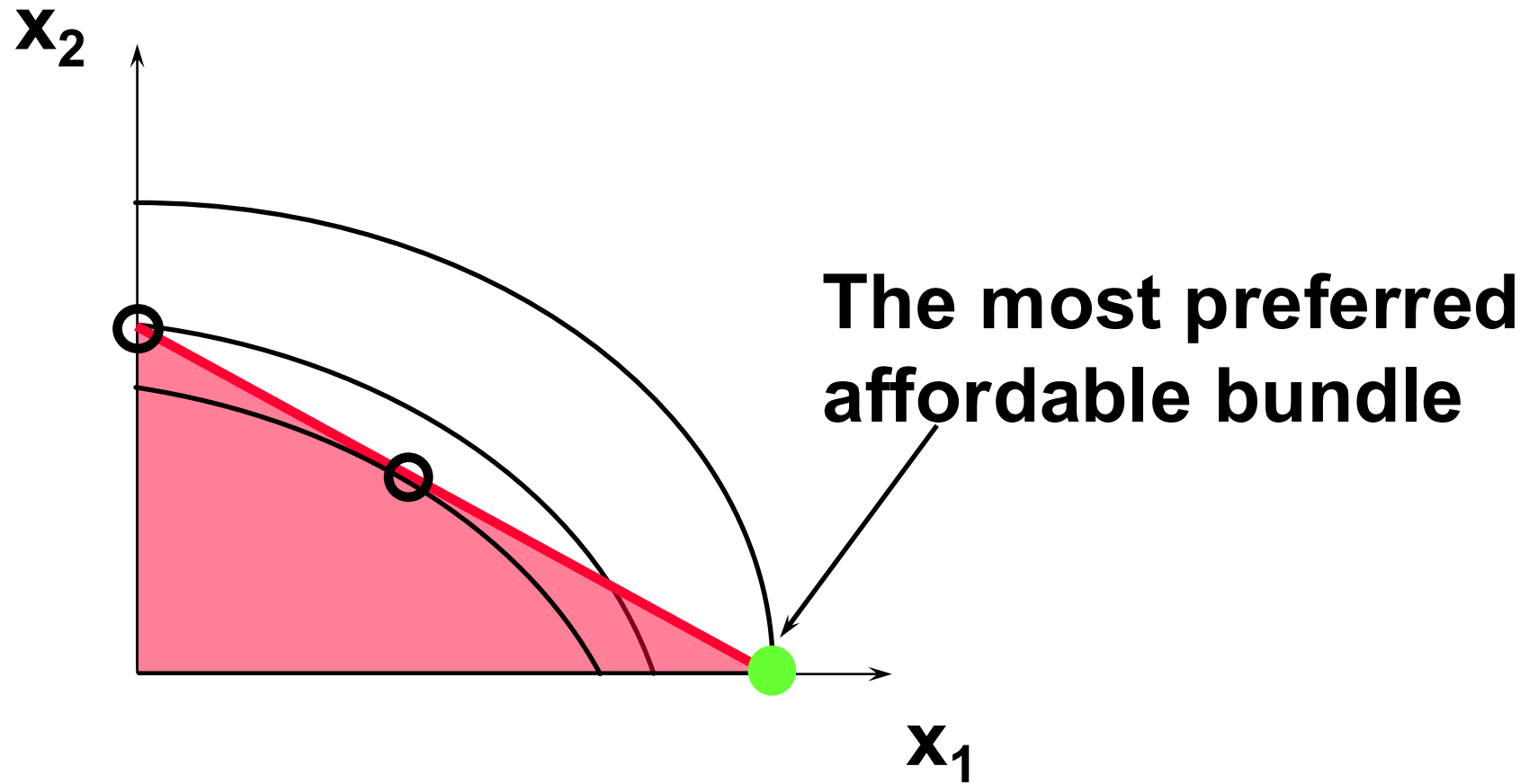
Examples of Corner Solutions

-- the Non-Convex Preferences Case



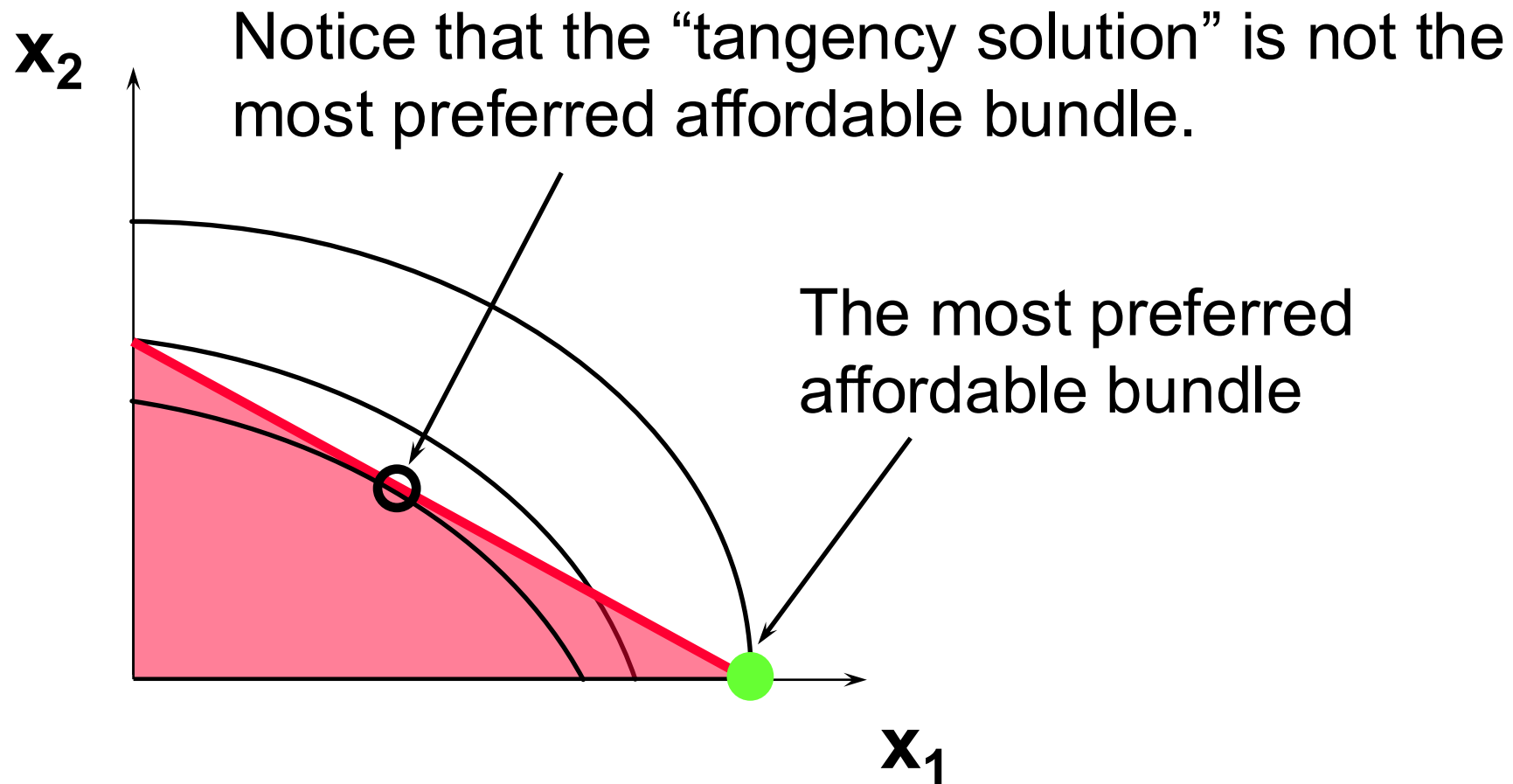
Examples of Corner Solutions

-- the Non-Convex Preferences Case



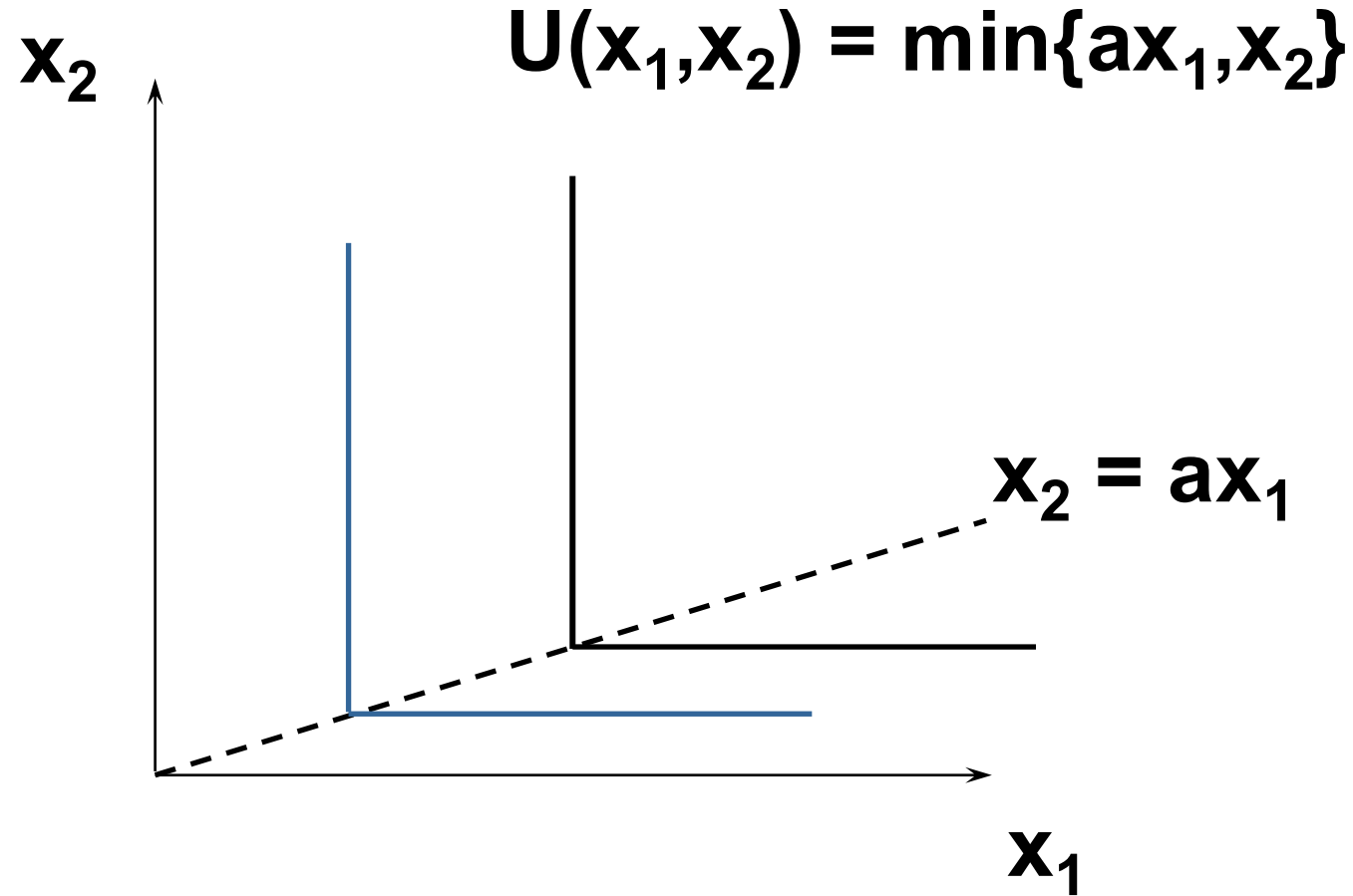
Examples of Corner Solutions

-- the Non-Convex Preferences Case



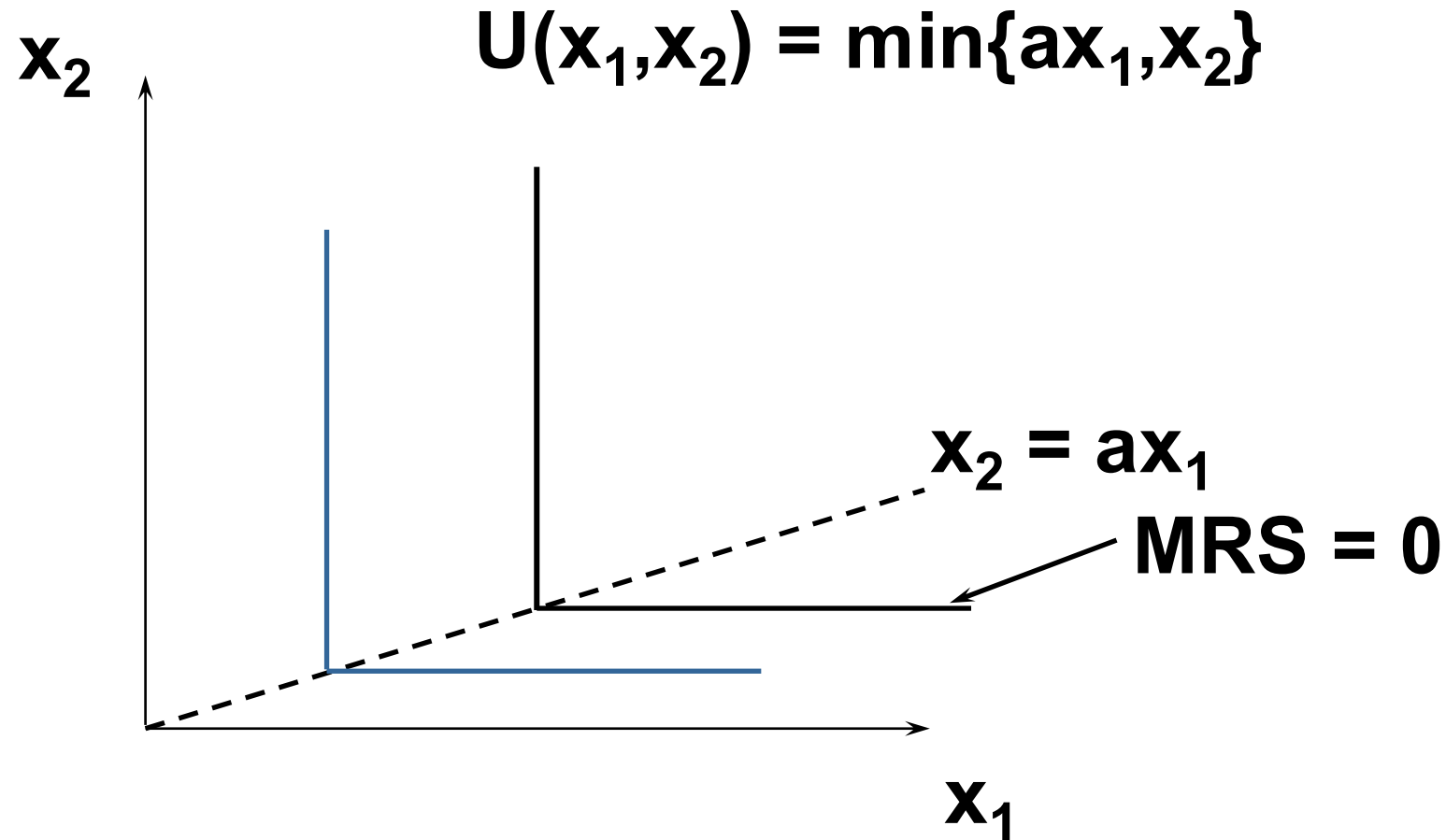
Examples of 'Kinky' Solutions

-- the Perfect Complements Case



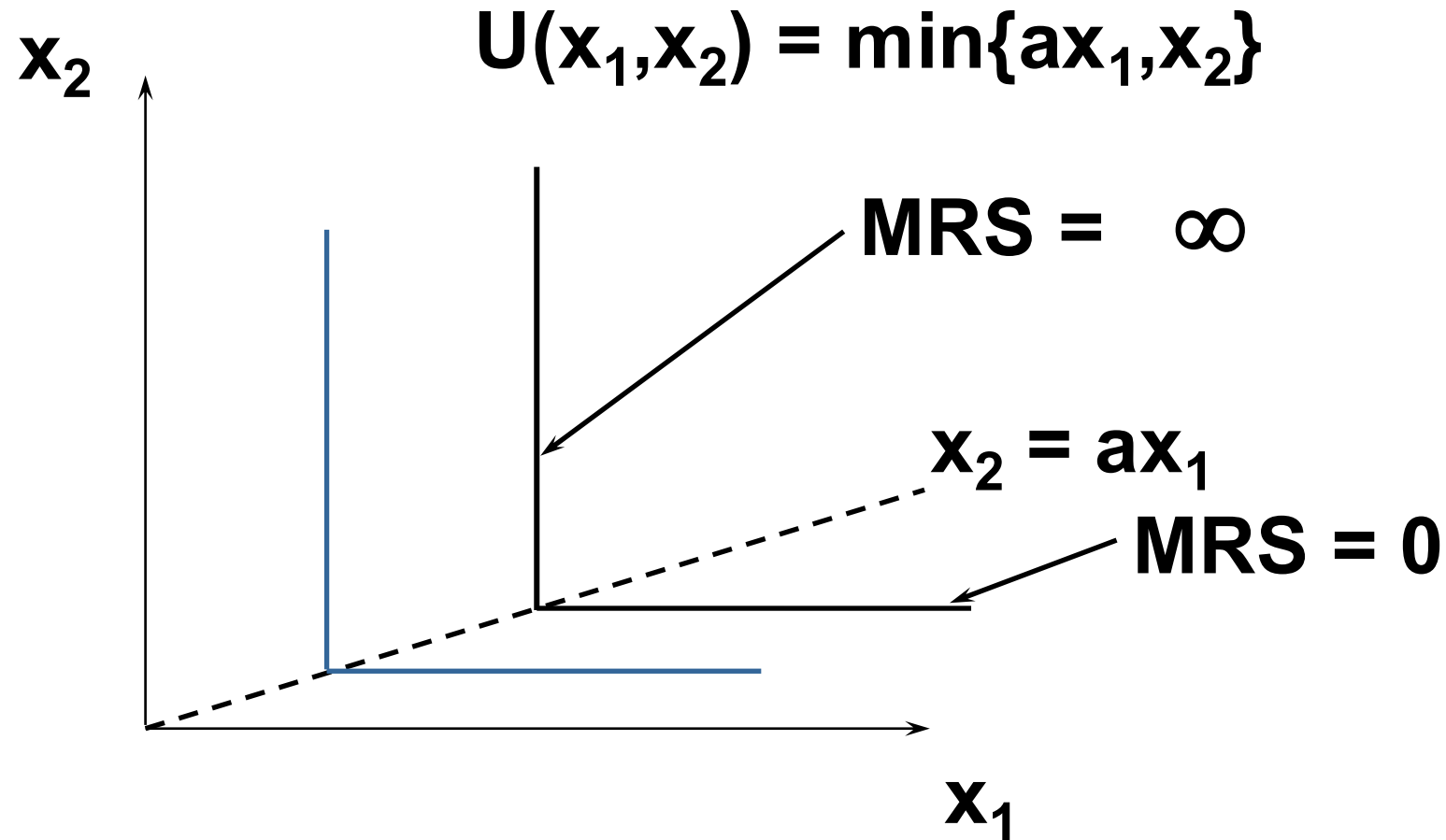
Examples of 'Kinky' Solutions

-- the Perfect Complements Case



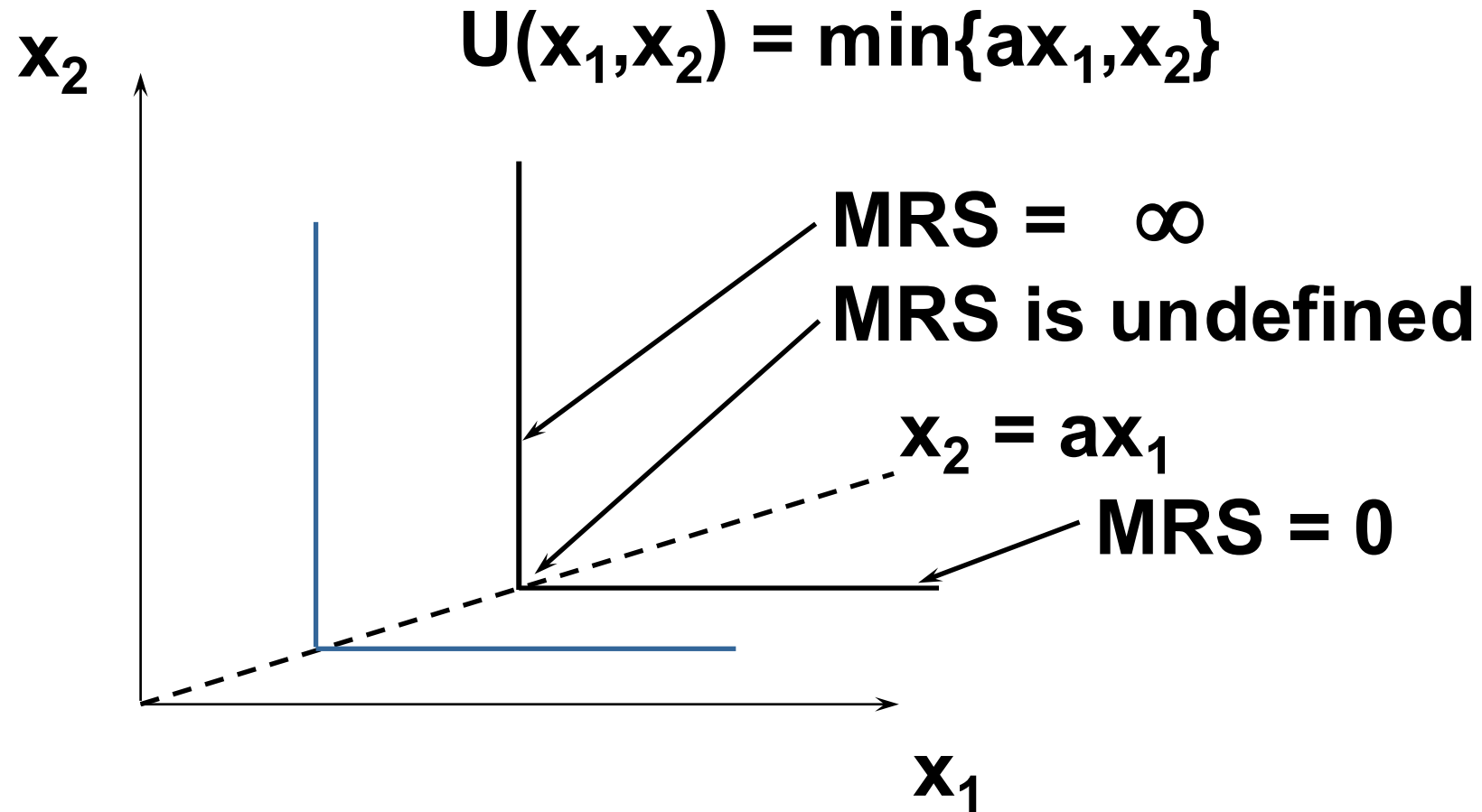
Examples of 'Kinky' Solutions

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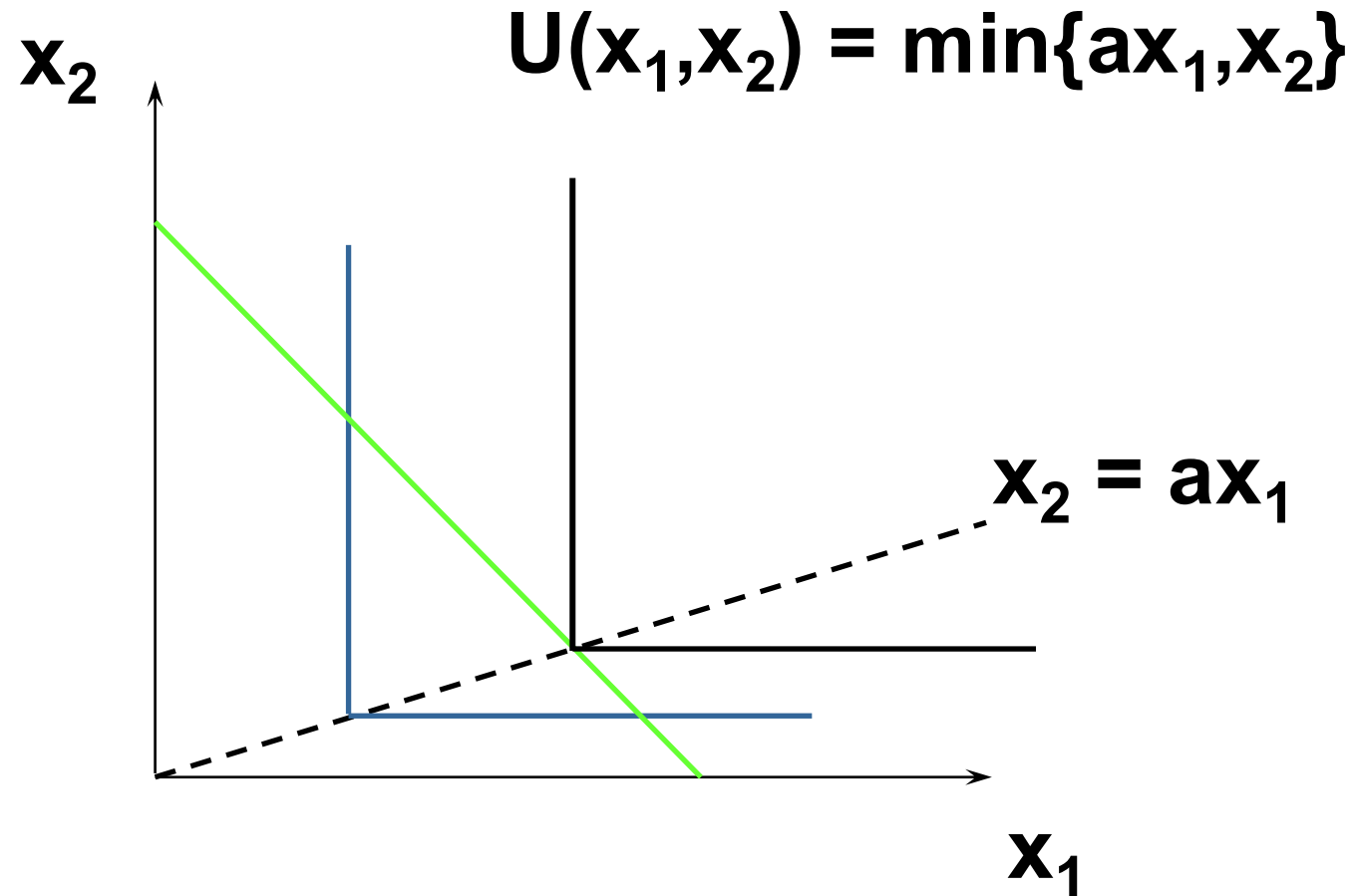
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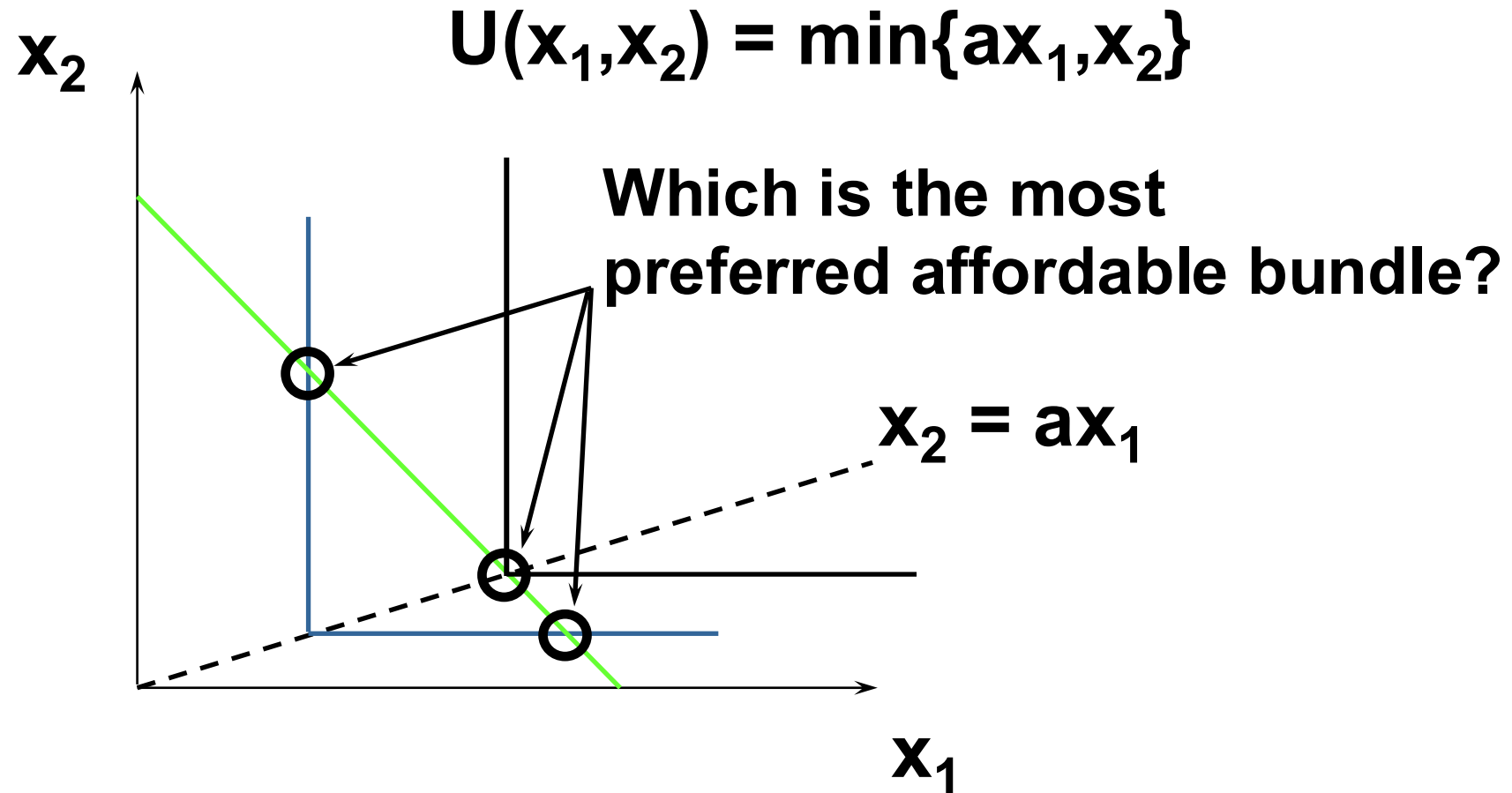
Examples of 'Kinky' Solutions

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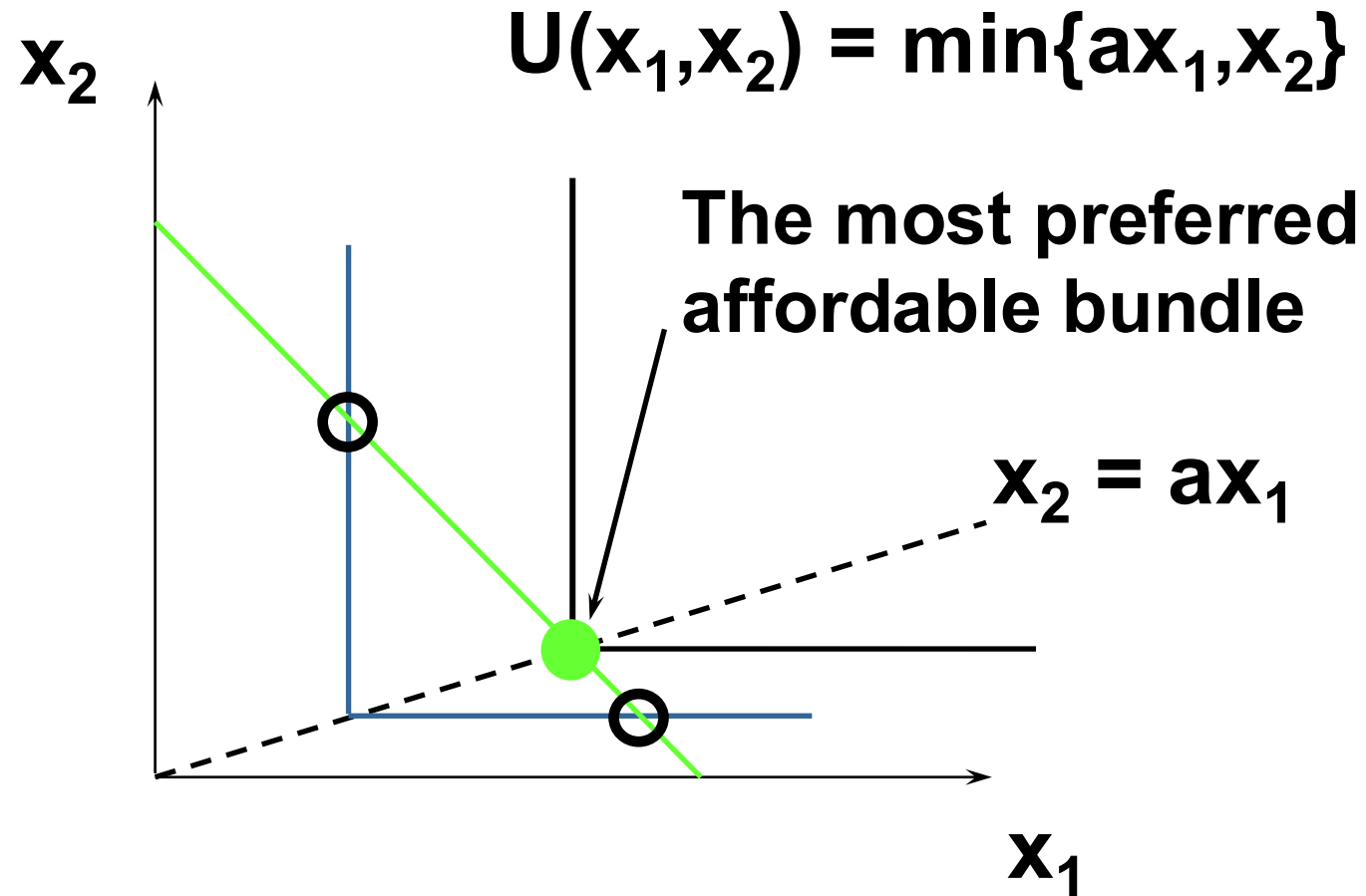
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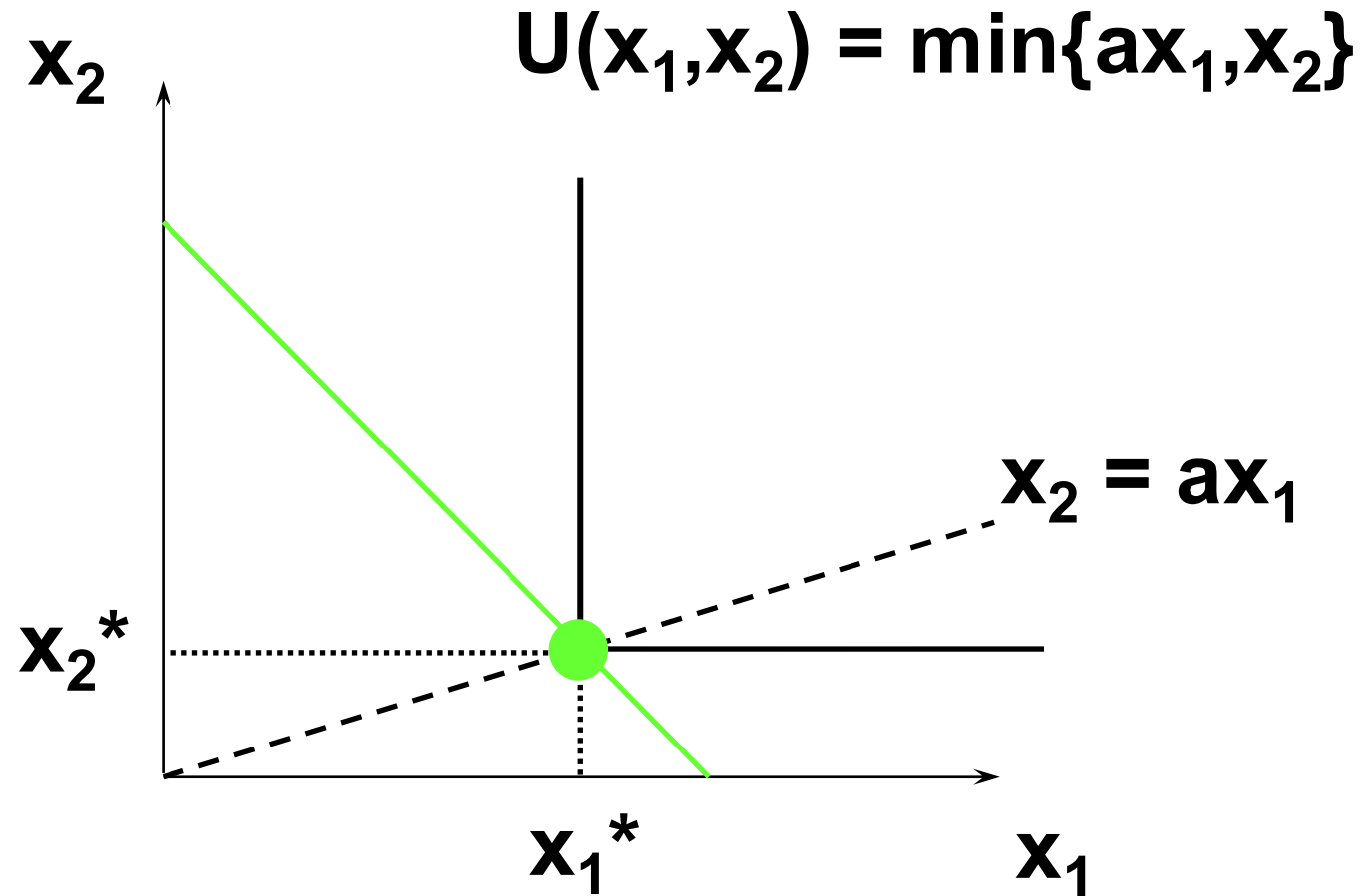
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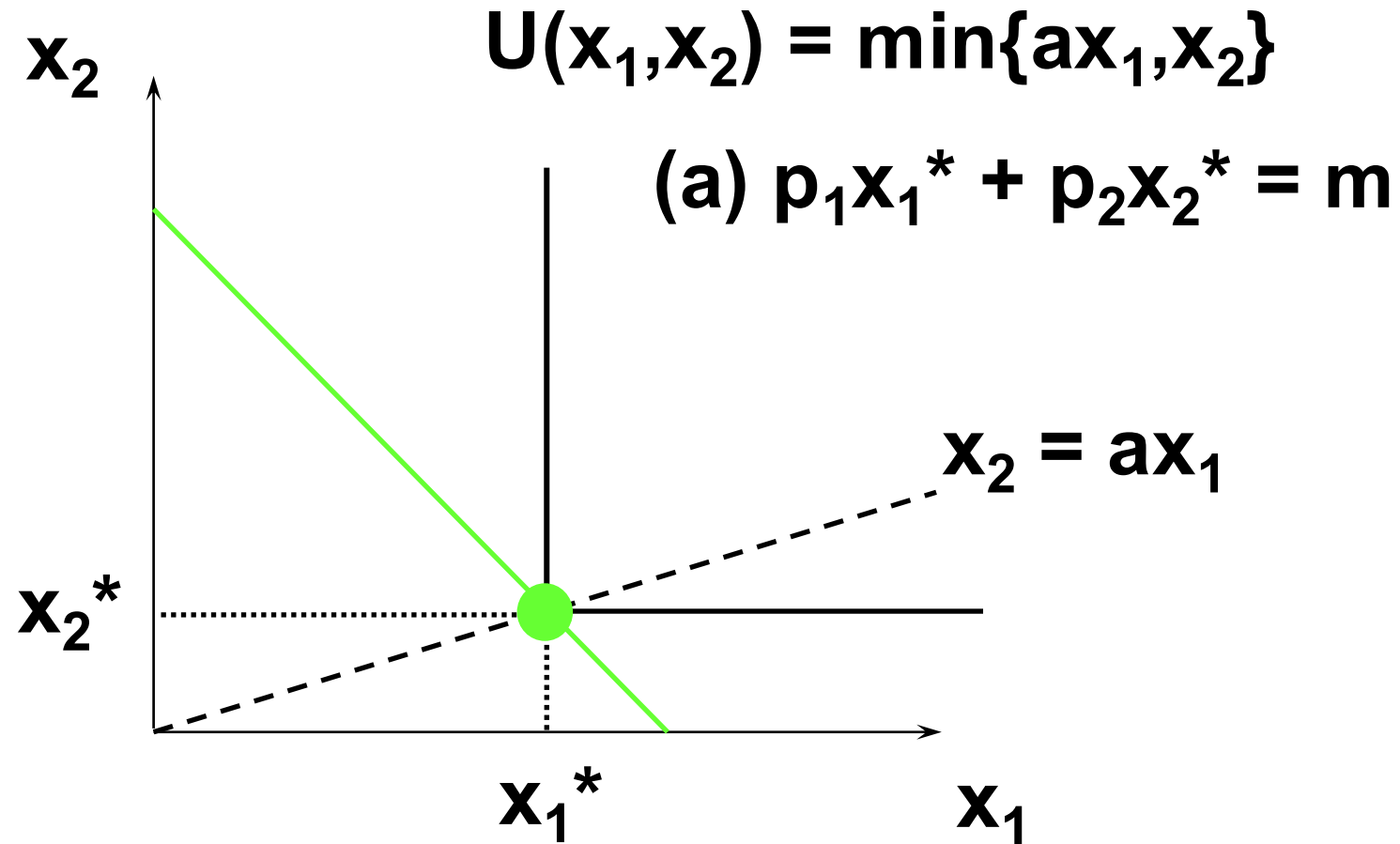
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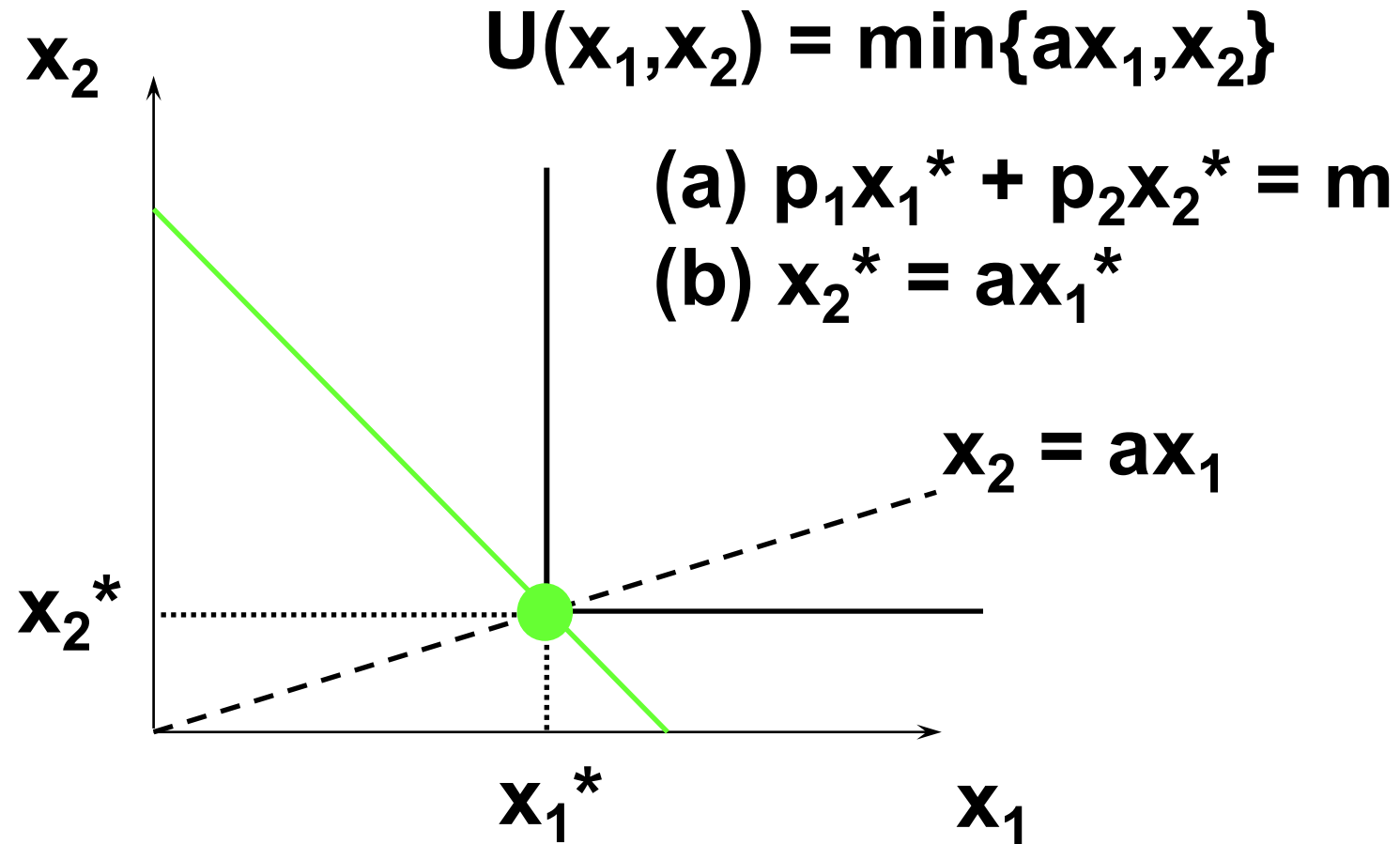
Examples of 'Kinky' Solutions

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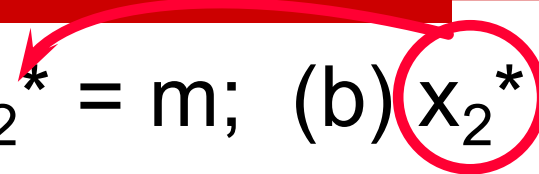
Examples of 'Kinky' Solutions

-- the Perfect Complements Case



Examples of 'Kinky' Solutions

-- the Perfect Complements Case

$$(a) p_1 x_1^* + p_2 x_2^* = m; \quad (b) x_2^* = a x_1^*.$$


Substitution from (b) for x_2^* in (a)
gives $p_1 x_1^* + p_2 a x_1^* = m$

Examples of 'Kinky' Solutions

-- the Perfect Complements Case

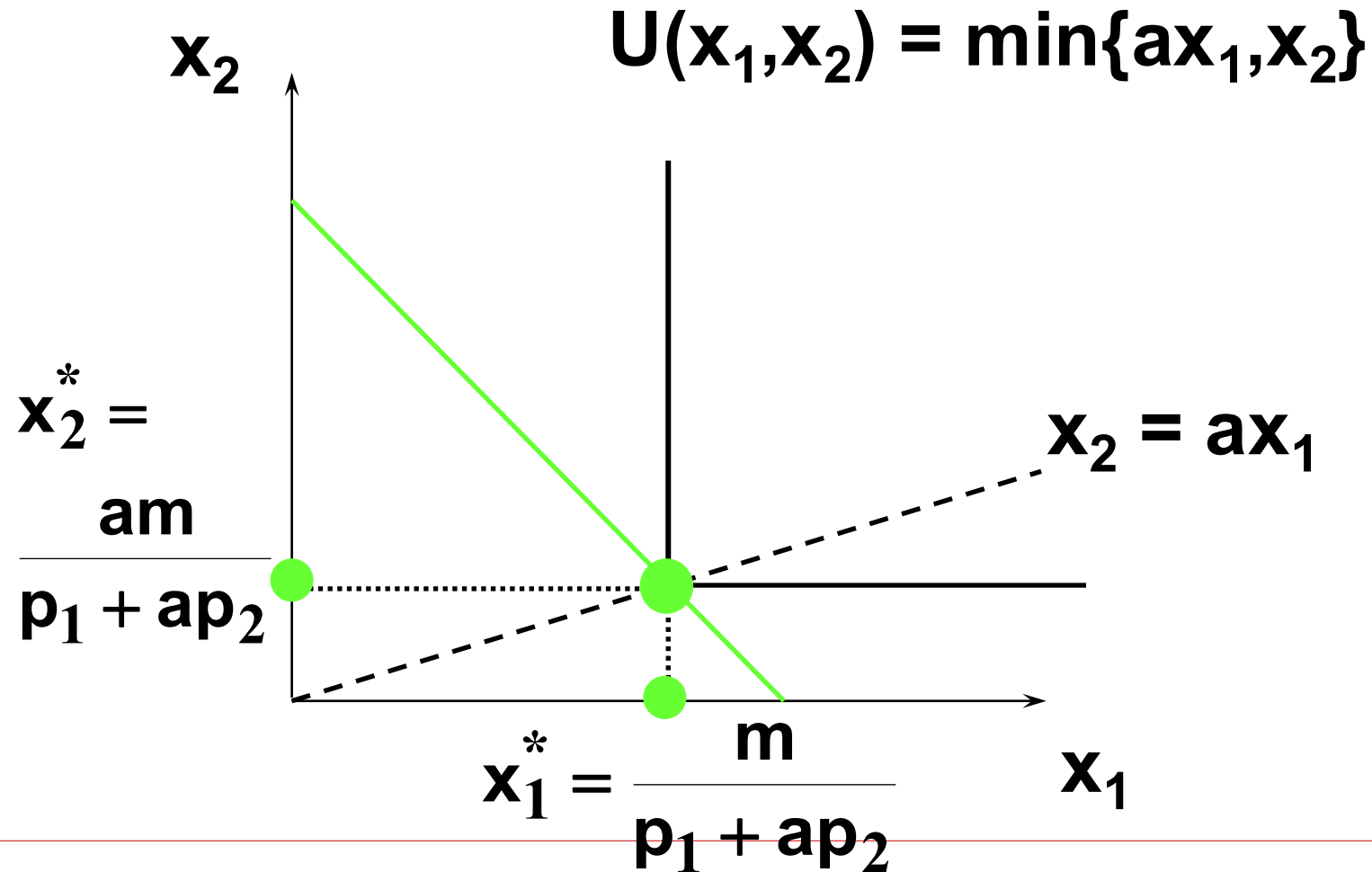
$$(a) p_1x_1^* + p_2x_2^* = m; \quad (b) x_2^* = ax_1^*.$$

Substitution from (b) for x_2^* in (a)
gives $p_1x_1^* + p_2ax_1^* = m$

which gives
$$\mathbf{x_1^* = \frac{m}{p_1 + ap_2}; \quad x_2^* = \frac{am}{p_1 + ap_2}.}$$

Examples of 'Kinky' Solutions

-- the Perfect Complements Case



The n -Good Case

- The individual's objective is to maximize

$$\text{utility} = U(x_1, x_2, \dots, x_n)$$

subject to the budget constraint

$$I = p_1x_1 + p_2x_2 + \dots + p_nx_n$$

- Set up the Lagrangian:

$$\mathbf{L} = U(x_1, x_2, \dots, x_n) + \lambda(I - p_1x_1 - p_2x_2 - \dots - p_nx_n)$$

The n -Good Case

□ First-order conditions for an interior maximum:

$$\partial L / \partial x_1 = \partial U / \partial x_1 - \lambda p_1 = 0$$

$$\partial L / \partial x_2 = \partial U / \partial x_2 - \lambda p_2 = 0$$

•
•
•

$$\partial L / \partial x_n = \partial U / \partial x_n - \lambda p_n = 0$$

$$\partial L / \partial \lambda = I - p_1 x_1 - p_2 x_2 - \dots - p_n x_n = 0$$

Implications of First-Order Conditions

□ For any two goods,

$$\frac{\partial U / \partial x_i}{\partial U / \partial x_j} = \frac{p_i}{p_j}$$

- This implies that at the optimal allocation of income

$$MRS (x_i \text{ for } x_j) = \frac{p_i}{p_j}$$

Interpreting the Lagrangian Multiplier

$$\lambda = \frac{\partial U / \partial x_1}{p_1} = \frac{\partial U / \partial x_2}{p_2} = \dots = \frac{\partial U / \partial x_n}{p_n}$$

$$\lambda = \frac{MU_{x_1}}{p_1} = \frac{MU_{x_2}}{p_2} = \dots = \frac{MU_{x_n}}{p_n}$$

- λ is the marginal utility of an extra dollar of consumption expenditure
 - the marginal utility of income

Cobb-Douglas Demand Functions

- Cobb-Douglas utility function:

$$U(x,y) = x^\alpha y^\beta, \quad \alpha + \beta = 1$$

- Setting up the Lagrangian:

$$\mathbf{L} = x^\alpha y^\beta + \lambda(I - p_x x - p_y y)$$

- First-order conditions:

$$\partial \mathbf{L} / \partial x = \alpha x^{\alpha-1} y^\beta - \lambda p_x = 0$$

$$\partial \mathbf{L} / \partial y = \beta x^\alpha y^{\beta-1} - \lambda p_y = 0$$

$$\partial \mathbf{L} / \partial \lambda = I - p_x x - p_y y = 0$$

Cobb-Douglas Demand Functions

- First-order conditions imply:

$$\alpha y / \beta x = p_x / p_y$$

- Since $\alpha + \beta = 1$:

$$p_y y = (\beta / \alpha) p_x x = [(1 - \alpha) / \alpha] p_x x$$

- Substituting into the budget constraint:

$$I = p_x x + [(1 - \alpha) / \alpha] p_x x = (1 / \alpha) p_x x$$

Cobb-Douglas Demand Functions

- Solving for x yields

$$x^* = \frac{\alpha I}{p_x}$$

Expenditure on x = $P_x * x$
Expenditure share of x =
 $P_x * x / I = \alpha$

- Solving for y yields

$$y^* = \frac{\beta I}{p_y}$$

Expenditure on y = $P_y * y$
Expenditure share of y =
 $P_y * y / I = \beta$

- The individual will allocate α percent of his income to good x and β percent of his income to good y

Cobb-Douglas Demand Functions

- The Cobb-Douglas utility function is limited in its ability to explain actual consumption behavior
 - the share of income devoted to particular goods often changes in response to changing economic conditions

- A more general functional form might be more useful in explaining consumption decisions

CES Demand

- Assume that $\delta = 0.5$

$$U(x,y) = x^{0.5} + y^{0.5}$$

$$I = p_x x + p_y y$$

- Solve for demand functions of x and y .

CES Demand

- In these demand functions, the share of income spent on either x or y is not a constant
 - depends on the ratio of the two prices
- The higher is the relative price of x (or y), the smaller will be the share of income spent on x (or y)

Perfect Complements

$$U(x,y) = \text{Min}(x,4y)$$

$$I = p_x x + p_y y$$

□ Solve for demand functions of x and y .

Indirect Utility Function

- It is often possible to manipulate first-order conditions to solve for optimal values of $x_1, x_2, \dots, x_n$
- These optimal values will depend on the prices of all goods and income

$$\begin{aligned}x_1^* &= x_1(p_1, p_2, \dots, p_n, I) \\x_2^* &= x_2(p_1, p_2, \dots, p_n, I) \\&\vdots \\x_n^* &= x_n(p_1, p_2, \dots, p_n, I)\end{aligned}$$

Indirect Utility Function

- We can use the optimal values of the x 's to find the indirect utility function

$$\text{maximum utility} = U(x^*_1, x^*_2, \dots, x^*_n)$$

- Substituting for each x^*_i , we get

$$\text{maximum utility} = V(p_1, p_2, \dots, p_n, I)$$

- The optimal level of utility will depend indirectly on prices and income
 - if either prices or income were to change, the maximum possible utility will change

Application

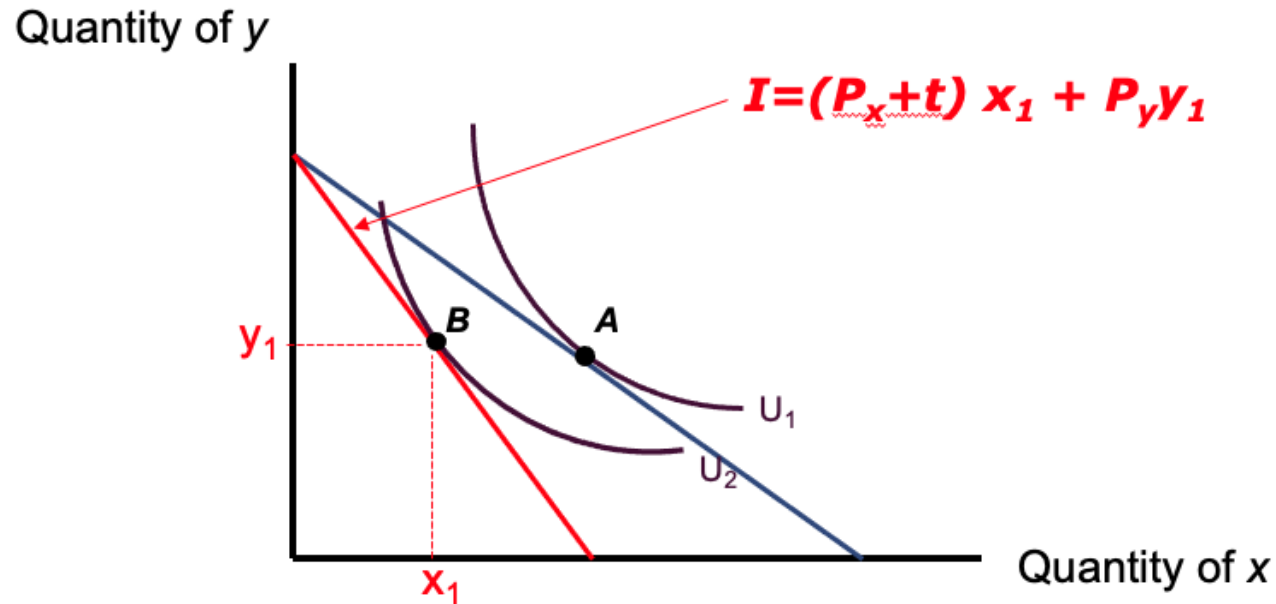
The Lump Sum Principle

- Taxes on an individual's general purchasing power are superior to taxes on a specific good

- From intuition:
 - an income tax allows the individual to decide freely how to allocate remaining income
 - a tax on a specific good will reduce an individual's purchasing power and distort his choices

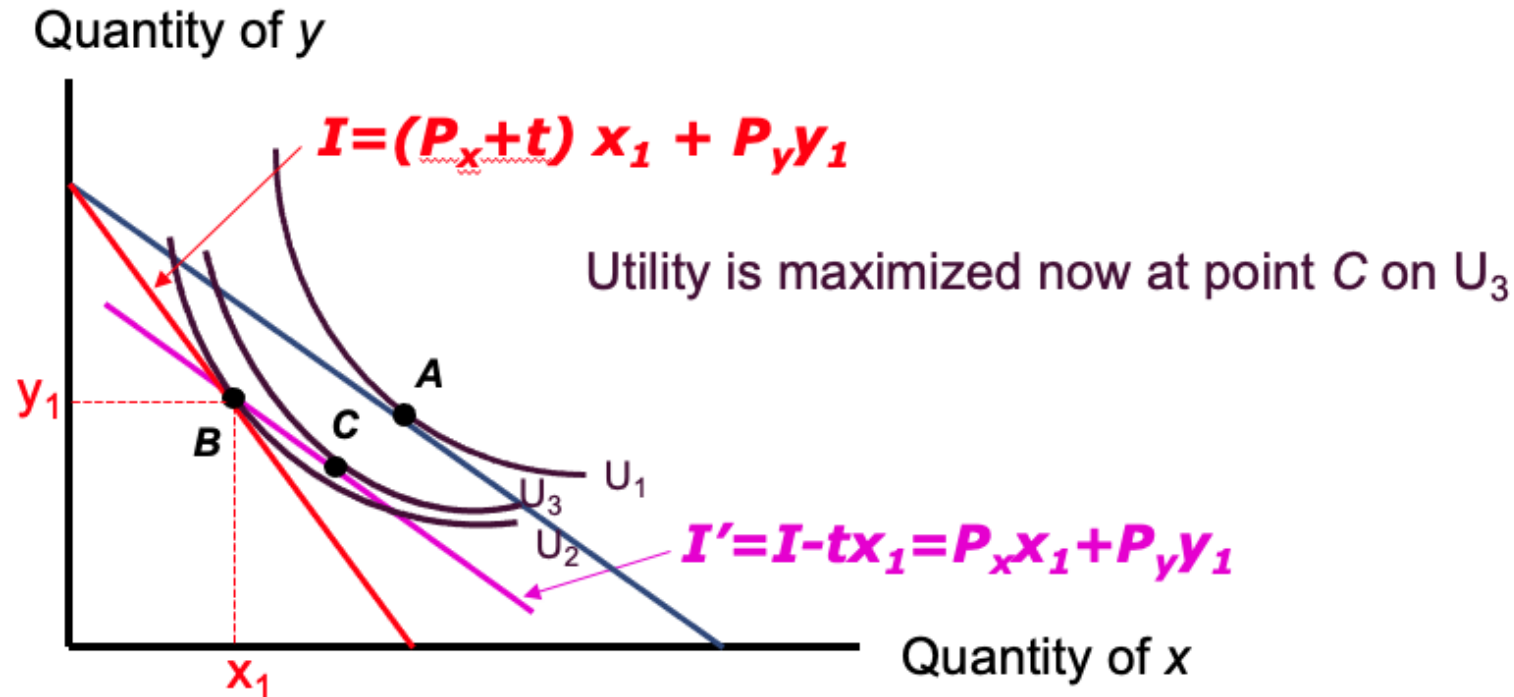
The Lump Sum Principle

- A tax on good x would shift the utility-maximizing choice from point A to point B



The Lump Sum Principle

- An income tax that collected the same amount would shift the budget constraint to I'



Indirect Utility and the Lump Sum Principle

- If the utility function is Cobb-Douglas with $\alpha = \beta = 0.5$, we know that

$$x^* = \frac{I}{2p_x} \qquad y^* = \frac{I}{2p_y}$$

So the indirect utility function is

$$V(p_x, p_y, I) = (x^*)^{0.5} (y^*)^{0.5} = \frac{I}{2p_x^{0.5} p_y^{0.5}}$$

If $p_x = 1$, $p_y = 4$, $I = 8$, then $V = \frac{8}{2 \cdot 1 \cdot 2} = 2$

Indirect Utility and the Lump Sum Principle

$$x^* = \frac{I}{2p_x} \quad y^* = \frac{I}{2p_y} \quad p_x = 1, \quad p_y = 4, \quad I = 8,$$

$$V(p_x, p_y, I) = (x^*)^{0.5} (y^*)^{0.5} = \frac{I}{2p_x^{0.5} p_y^{0.5}}$$

Then $V = 2$.

- If a tax of \$1 per unit was imposed on good x
 - the individual will purchase $x^*=2$
 - indirect utility will fall from 2 to 1.41
- An equal-revenue tax will reduce income to \$6
 - indirect utility will fall from 2 to 1.5

Another example -- Perfect complements

- If the utility function is fixed proportions with $U = \text{Min}(x, 4y)$, we know that

$$x^* = \frac{I}{p_x + 0.25p_y} \qquad y^* = \frac{I}{4p_x + p_y}$$

So the indirect utility function is

$$\begin{aligned} V(p_x, p_y, I) &= \text{Min}(x^*, 4y^*) = x^* = \frac{I}{p_x + 0.25p_y} \\ &= 4y^* = \frac{4I}{4p_x + p_y} = \frac{I}{p_x + 0.25p_y} \end{aligned}$$

Indirect Utility and the Lump Sum Principle

$$p_x = 1, p_y = 4, I = 8$$

- If a tax of \$1 per unit was imposed on good x
 - indirect utility will fall from 4 to $8/3$

- An equal-revenue tax will reduce income to $\$16/3$
 - indirect utility will fall from 4 to $8/3$

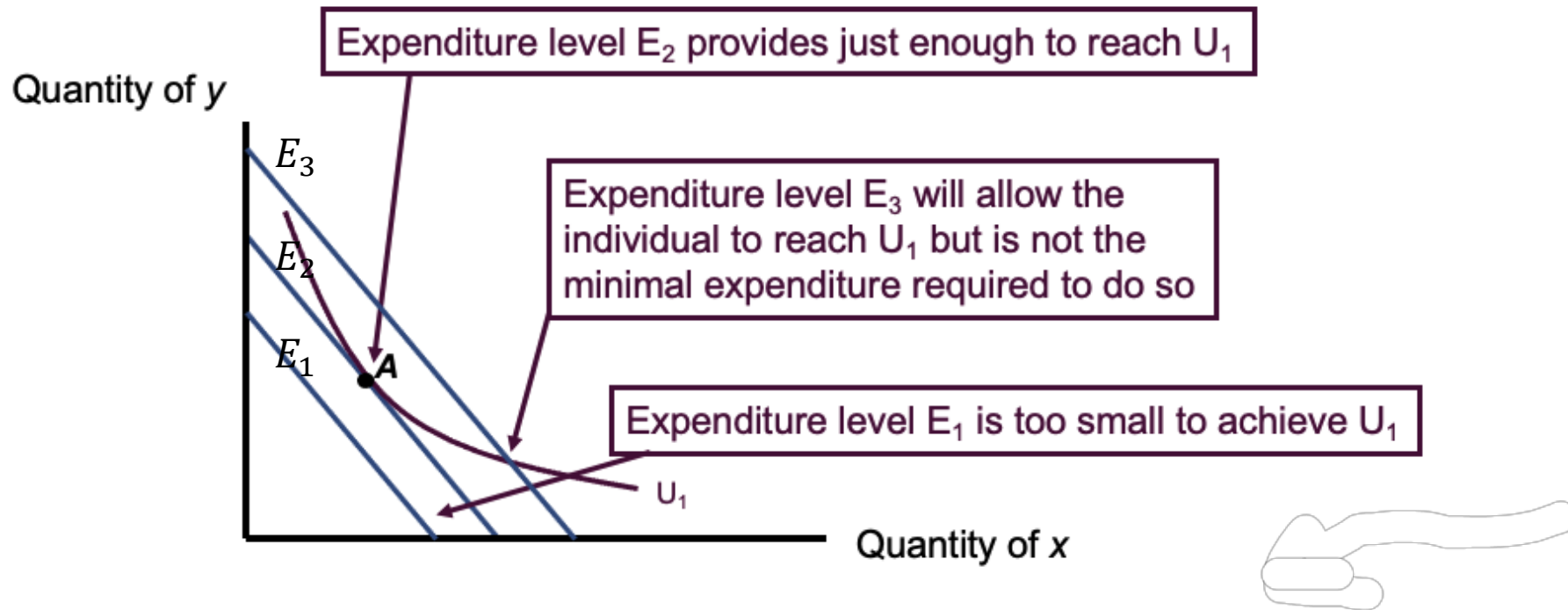
- Since preferences are rigid, the tax on x does not distort choices

Expenditure Minimization

- Dual minimization problem for utility maximization
 - allocating income in such a way as to achieve a given level of utility with the minimal expenditure
 - this means that the goal and the constraint have been reversed

Expenditure Minimization

Point A is the solution to the dual



Expenditure Minimization

- The individual's problem is to choose $x_1, x_2, \dots, x_n$ to minimize

$$\text{total expenditures} = E = p_1x_1 + p_2x_2 + \dots + p_nx_n$$

subject to the constraint

$$\text{utility} = U_1 = U(x_1, x_2, \dots, x_n)$$

- The optimal amounts of $x_1, x_2, \dots, x_n$ will depend on the prices of the goods p_i 's and the required utility level U_1 .

Expenditure Function

- The expenditure function shows the minimal expenditures necessary to achieve a given utility level for a particular set of prices

$$\text{minimal expenditures} = E(p_1, p_2, \dots, p_n, U_1)$$

Expenditure Minimization

□ Construct the Lagrangian

$$L = p_1x_1 + p_2x_2 + \cdots + p_nx_n + \lambda(U_0 - U(x_1, x_2, \cdots, x_n))$$

□ FOCs:

$$\partial L / \partial x_1 = p_1 - \lambda \partial U / \partial x_1 = 0$$

$$\partial L / \partial x_2 = p_2 - \lambda \partial U / \partial x_2 = 0$$

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$$\partial L / \partial x_n = p_n - \lambda \partial U / \partial x_n = 0$$

$$\partial L / \partial \lambda = U_0 - U(x_1, x_2, \cdots, x_n) = 0$$

Expenditure Minimization

The diagram shows the equation $\lambda = \frac{p_1}{\partial U / \partial x_1} = \frac{p_2}{\partial U / \partial x_2} = \dots = \frac{p_n}{\partial U / \partial x_n}$. A box labeled "MC of 1 more unit of x_i " has an arrow pointing to the p_1 term in the numerator. Another box labeled "MU of 1 more unit of x_i " has an arrow pointing to the $\partial U / \partial x_1$ term in the denominator.

$$\lambda = \frac{p_1}{\partial U / \partial x_1} = \frac{p_2}{\partial U / \partial x_2} = \dots = \frac{p_n}{\partial U / \partial x_n}$$

- In expenditure minimization problem:
 - λ is the marginal expenditure required to reach an extra unit of utility
 - In utility maximization problem:
 - λ is the marginal utility of an extra dollar of consumption expenditure
-

How are indirect utility function and expenditure function related?

- The indirect utility function in the two-good, Cobb-Douglas case is

$$V(p_x, p_y, I) = \frac{I}{2p_x^{0.5} p_y^{0.5}}$$

If we interchange the role of utility and income (expenditure), we will have the expenditure function

$$E(p_x, p_y, U) = 2p_x^{0.5} p_y^{0.5} U$$

How are indirect utility function and expenditure function related?

- For the fixed-proportions case, the indirect utility function is

$$V(p_x, p_y, I) = \frac{I}{p_x + 0.25p_y}$$

If we again switch the role of utility and expenditures, we will have the expenditure function

$$E(p_x, p_y, U) = (p_x + 0.25p_y)U$$

- The expenditure function and the indirect utility function are **inversely related**